

Agent Defense in Abstract Argumentation: Semantics and Principle-Based Analysis

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Abstract.

Dung's theory of abstract argumentation is a fundamental framework for knowledge representation and reasoning. It models the acceptability of arguments through their attack and defense relations, a perspective referred to as the attack-defense paradigm shift. While formal argumentation has been conceptualized in terms of argumentation as inference, argumentation as dialogue, and argumentation as balancing, most developments in abstract argumentation have focused on the inference perspective. By contrast, the dialogue perspective remains comparatively underexplored. In this article, we contribute to bridging this gap by introducing new notions of agent defense, extending abstract argumentation with explicit representations of agents and their roles in defending arguments. These notions account for both individual and collective defense, enabling richer models of multi-agent reasoning. We position our proposal within the literature by comparing it with three existing approaches that extend abstract argumentation with agency: social semantics, agent-reduction semantics, and agent-filtering semantics. Using a principle-based analysis, we evaluate the formal properties of these approaches and the behavioral differences between them. This work broadens the focus of abstract argumentation from inference-oriented models toward dialogue-oriented and agent-centered perspectives. This aligns with ongoing developments described in the Handbook of Formal Argumentation and the International Conference on Computational Models of Argument (COMMA) literature, and contributes to the shift toward modeling complex, interactive reasoning in multi-agent systems.

Keywords: Knowledge representation and reasoning, abstract agent argumentation, principle-based approach

1. Introduction

Dung's theory of abstract argumentation [22] provides a unified foundation for different approaches to knowledge representation and reasoning in the field of artificial intelligence. The core concept of abstract argumentation is *attack*: the theory abstracts from the internal structure of arguments in order to focus on the attack relations between them. *Defense* is defined in terms of attack: an argument is defended by a set of arguments if every attacker of it is attacked by at least one argument in the set. The concept of defense further determines the acceptability or non-acceptability of arguments. In this regard, Dung's abstract argumentation is seen as the attack-defense paradigm shift of formal argumentation [50].

The first volume of the Handbook of Formal Argumentation [9] explains the central role of Dung's theory of abstract argumentation [22], while the second volume [9] presents many variants proposed over the past few decades such as bipolar argumentation, preference argumentation framework, etc. Our article is inspired by an argumentation-as-dialogue approach that is based on agents, strategies, and games. We extend Dung's abstract argumentation framework with a set of agents. Agent-based extensions typically introduce various aspects such as knowledge, uncertainty, support, trust, and so on. We

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treat a minimal extension of Dung's theory [22] as the common core of these approaches. That minimal extension is limited to an abstract set of agents, and all arguments are associated with agents. Although *agent-based* variants have been proposed in several studies [6, 25, 30], an overview of these variants is still lacking. Moreover, the semantics of agent argumentation concerns the merging of argumentation frameworks [17, 20, 21].

This article is focused on adapting Dung's notion of defense within agent-based argumentation frameworks. It introduces two new types of agent-specific defenses: individual defense and collective defense. These novel semantics modify the traditional attack-defense structure by taking into consideration the agents associated with each argument, which adds new dimensions to the concept of admissibility.

We address the following research questions:

- (1) How to adapt the concept of defense in abstract agent argumentation frameworks?
- (2) How do the new concepts of agent defense compare to those of other approaches to abstract argumentation semantics?

Below, we compare the defense-based semantics with other traditional approaches to abstract agent argumentation.

Social approaches [32] are often inspired by concepts from social choice and voting theory. They give preference to arguments or attacks associated with more than one agent [6, 7, 17, 30]. They integrate argument preference orderings [1] by counting the number of agents that hold these arguments and ranking the arguments accordingly. This kind of approach shares some attributes with the voting mechanism employed in Social Choice Theory [5] which provides the classical means of aggregating individual agent preferences into collective decisions [30].

Reduction-based approaches are motivated by judgment aggregation in Social Choice Theory [26, 35], which considers the perspectives of each agent individually. There are two different approaches to collective acceptability in the literature — the argument-wise approach and the framework-wise approach [14]. The argument-wise approach aggregates individually-accepted arguments into a single collectively-acceptable set of arguments. The framework-wise approach first defines structural aggregation methods to aggregate individual views into a collective representation, then determines the acceptability of arguments in the collective framework.

Filtering methods emphasize agents' knowledge or trust [4]. Each agent can only present the arguments and attacks they know and believe to be true [4]. Knowledge and belief are crucial elements for the argumentation framework. It filters out any argument or attack that is not associated with any agent.

Principles and axioms can be used in many ways. Often, they conceptualize the behavior of a system at a higher level of abstraction. Moreover, in the absence of a standard approach, principles can be used as a guide to choose the most appropriate definitions and semantics in accordance with particular requirements. Therefore, in formal argumentation, principles are often more technical. The principles most discussed in the literature are admissibility, directionality, and strongly-connected-component (SCC) decomposability. These principles play a central role in this article, especially those principles that enable us to distinguish between different kinds of agent semantics. Our work also fits naturally within the argumentation as balancing, dialogue and inference (A-BDI) metamodel [49] for formal and computational argumentation, which emphasizes the need for principled methodologies to connect diverse approaches and guide application-driven choices in artificial intelligence.

This article is a major expansion of the work we presented [47] at the 18th International Conference on Principles of Knowledge Representation and Reasoning (KR 2021). Specifically, we:

- (1) correct several errors in the result tables on principle-based analysis;
- (2) refine the treatment of *stable* semantics, clarifying that in our setting, the choice between individual and collective defense does not affect stable semantics;
- (3) introduce a new class of *argument-wise agent reduction* alongside the existing *framework-wise agent reduction* class;
- (4) provide full proofs and counterexamples for all stated results; and
- (5) substantially extend the Related Work and Future Work sections.

The article is organized as follows. Section 2 introduces agent argumentation frameworks, and it extends Dung’s abstract argumentation framework by associating arguments with agents. Section 3 describes two new notions of agent defense — individual defense and collective defense — and analyzes their respective roles in admissibility and reinstatement. Section 4 compares these defense-based semantics with three existing approaches: social agent semantics, agent reduction semantics, and agent filtering semantics. Section 5 examines the role of principles in evaluating agent argumentation semantics, with particular emphasis on admissibility, directionality, SCC-recursiveness, and other key principles. Section 6 introduces additional agent-based argumentation principles to facilitate finer-grained comparison of different semantics. Section 7 discusses related work, situating our approach within the broader literature on argumentation frameworks and their agent-based extensions. Section 8 outlines future directions, which may include integrating agent argumentation with structured argumentation and dialogue models. Section 9 concludes the article with a summary of our main findings and their implications for agent-based reasoning.

2. Agent Argumentation Framework

Agent argumentation frameworks generalize the argumentation frameworks studied by Dung (1995), which are directed graphs in which nodes represent arguments and arrows represent the attack relation.

Definition 1 (Argumentation framework [22]). An *argumentation framework* (AF) is a pair $\langle \mathcal{A}, \rightarrow \rangle$ where $\mathcal{A}$ is a set called arguments and $\rightarrow \subseteq \mathcal{A} \times \mathcal{A}$ is a binary relation over $\mathcal{A}$ called attack. For a set $S \subseteq \mathcal{A}$ and an argument $a \in \mathcal{A}$, we say that S attacks a if there exists a $b \in S$ such that b attacks a . Similarly, we say that a attacks S if there exists a $b \in S$ such that a attacks b . We define $a^- = \{b \in \mathcal{A} | b \text{ attacks } a\}$ and $S_{out}^- = \{a \in \mathcal{A} \setminus S | a \text{ attacks } S\}$.

Dung’s admissibility-based semantics is based on the concept of defense.

Definition 2 (Admissible [22]). Let $\langle \mathcal{A}, \rightarrow \rangle$ be an AF. A set $E \subseteq \mathcal{A}$ is *conflict-free* iff there are no arguments a and b in E such that a attacks b . We say that E *defends* an argument c iff for all arguments b attacking c , there is an argument $a \in E$ such that a attacks b . Finally, E is *admissible* iff it is conflict-free and defends all its elements.

For their principle-based analysis, Baroni and Giacomin [10] define semantics as a function that maps argumentation frameworks to a set of extensions (each extension being a subset of acceptable arguments).

Definition 3 (Dung's semantics [10]). Dung's semantics is a function σ that associates an argumentation framework $AF = \langle \mathcal{A}, \rightarrow \rangle$ with a set of subsets of $\mathcal{A}$. The elements of $\sigma(AF)$ are called extensions.

Dung introduced definitions for different types of extensions.

Definition 4 (Extensions [22]). Let $\langle \mathcal{A}, \rightarrow \rangle$ be an AF. A set $E \subseteq \mathcal{A}$ is a *complete extension* iff it is admissible and contains all the arguments it defends. $E \subseteq \mathcal{A}$ is a *grounded extension* iff it is the smallest complete extension (under set inclusion). $E \subseteq \mathcal{A}$ is a *preferred extension* iff it is a maximal complete extension (under set inclusion). $E \subseteq \mathcal{A}$ is a *stable extension* iff it is conflict-free and attacks every argument that is not in E .

Each type of extension can be seen as an acceptability semantics that formally rules the argument evaluation process. In this article, we use $\sigma \in \{c, g, p, s\}$ to represent Dung semantics {complete, grounded, preferred, stable}.

Example 1 (Two conflicts). Consider the argumentation framework visualized on the left in Figure 1, where $\mathcal{A} = \{a, b, c, d\}$ and $\rightarrow = \{a \rightarrow b, b \rightarrow a, c \rightarrow d, d \rightarrow c\}$. Each argument defends itself. There are nine admissible sets — $\{a\}, \{b\}, \{c\}, \{d\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \emptyset$ — which are all complete extensions. The grounded extension is $\emptyset$. The preferred extensions $\{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}$ are also stable extensions. For example, in an oft-used dinner scenario, we may choose between eating (a) fish or (b) meat, between (c) eating at home or (d) going out (d). These two choices are independent of one another. In structured argumentation, these arguments can have a complex structure in which the reasons behind conclusions are provided. However, in abstract argumentation, the reasons are not specified.

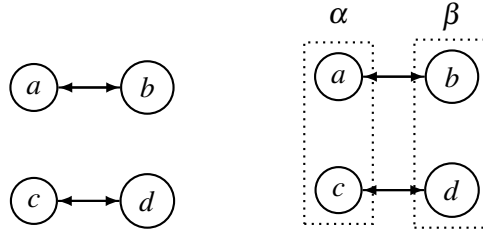


Fig. 1. An agent framework (AF) and an agent argument framework (AAF)

An agent argumentation framework extends an argumentation framework with a set of agents and a relation that associates arguments with agents. Note that an argument can belong to no agent, one agent, or multiple agents. This is the most general case. We briefly discuss restrictions in the further work section toward the end of this article.

We write $a \sqsubset \alpha$ to say that argument a belongs to agent α , or that agent α has argument a .

Definition 5 (Agent argumentation framework). An *agent argumentation framework* (AAF) is a 4-tuple $\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ where $\mathcal{A}$ is a finite set of arguments, $\rightarrow \subseteq \mathcal{A} \times \mathcal{A}$ is a binary relation over $\mathcal{A}$ called attack, $\mathcal{S}$ is a set of agents or sources, and $\sqsubset \subseteq \mathcal{A} \times \mathcal{S}$ is a binary relation associating arguments with agents. For each agent α , we write $\mathcal{A}_\alpha = \{a \in \mathcal{A} | a \sqsubset \alpha\}$ for the set of all arguments that belong to agent α . For each argument a , we write $\mathcal{S}_a = \{\alpha | a \sqsubset \alpha\}$ for the set of all agents that have argument a , and we write $\rightarrow_a = \{x \rightarrow y | x = a \text{ or } y = a\}$ for the set of attack relations involving argument a . Finally, for each agent α , we write $\sqsubset_\alpha = \{(a, \alpha) | a \sqsubset \alpha\}$, for the binary relation that links α to its arguments.

Example 2 (Two conflicts, continued from Example 1). Consider the agent argumentation framework visualized on the right in Figure 1. The figure should be read as follows. Each dashed box contains all the arguments held by the agents in $S = \{\alpha, \beta\}$; the argument-agent relation is $\sqsubset = \{(a, \alpha), (b, \beta), (c, \alpha), (d, \beta)\}$. For example, Alice (α) argues in favor of eating fish and staying at home while Bob (β) argues in favor of eating meat and going out.

3. Agent Defense Semantics

We now introduce a new kind of defense for agent argumentation frameworks called agent defense. Basically, if an agent puts forward an argument, it can only be defended by arguments held by that same agent. This reflects coalition-based reasoning as discussed by Qiao et al. [39].

In individual agent defense, only one agent can defend each argument, whereas in collective agent defense, a set of agents can defend each argument.

Definition 6 (Agent Admissible). Let $\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be an AAF:

- $E \subseteq \mathcal{A}$ is *conflict-free* iff there are no arguments a and b in E such that a attacks b .
- $E \subseteq \mathcal{A}$ *individually agent-defends* (agent-defends_1) c iff there exists an agent α in $\mathcal{S}_c$ such that for all arguments b in $\mathcal{A}$ attacking c , there exists an argument a in $E \cap \mathcal{A}_\alpha$ such that a attacks b .
- $E \subseteq \mathcal{A}$ *collectively agent-defends* (agent-defends_2) c iff for all arguments b in $\mathcal{A}$ attacking c , there exists an agent α in $\mathcal{S}_c$ and an argument a in $E \cap \mathcal{A}_\alpha$ such that a attacks b .
- $E \subseteq \mathcal{A}$ is *agent admissible_i* iff it is conflict-free and agent-defends_i all its elements, for i in $\{1, 2\}$.

The following example illustrates *individual agent defense* and its role in the well-known property of *reinstatement* [38]. The ultimate status of an argument depends on the interaction between all available arguments. It may very well be that argument a attacks argument b but that a is itself attacked by an argument c . If no argument successfully attacks c , then c *reinstates* b . While reinstatement is considered by many to be a desirable property, others are of the opinion that it should not hold in general; see for instance Horty [27]. Example 3 shows that there is a middle way in this debate. Under individual agent defense, reinstatement is only allowed if the argument being defended and all the arguments that defend it are held by the same agent. If the defenders belong to different agents, reinstatement does not occur.

Example 3 (Reinstatement). Consider the agent argumentation framework in Figure 2, where $\mathcal{A} = \{a, b, c\}$, and $\rightarrow = \{c \rightarrow b, b \rightarrow a\}$, and $\mathcal{S} = \{\alpha, \beta, \gamma\}$, and $\sqsubset = \{(a, \alpha), (b, \beta), (c, \gamma)\}$. In this structure, argument c *defends* argument a by attacking its attacker b , but it does **not** *individually agent-defend* a , because individual agent defense requires that all a 's defenders should come from the agent holding a .

In the dinner scenario, Alice (α) argues for having meat (a), Bob (β) argues for having fish (b), and Cayrol (γ) argues against having fish with argument c . In the standard setting, Cayrol's attack on Bob would reinstate Alice's argument. But under individual agent defense, Alice cannot use Cayrol's argument to defend her own argument, so reinstatement does not occur.

Definition 7 (Agent semantics). An agent semantics is a function δ that associates a set of subsets of $\mathcal{A}$ with an agent argumentation framework $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$; the elements of $\delta(AAF)$ are called agent extensions.

We use Sem_1 to represent agent semantics based on individual defense and Sem_2 to represent agent semantics based on collective defense.

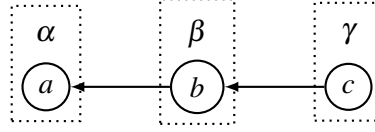


Fig. 2. Agent reinstatement

Definition 8 (Agent extensions). Let $\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be an agent argumentation framework (AAF) where:

- $E \subseteq \mathcal{A}$ is an *agent-complete_i extension* iff E is agent-admissible_i and contains all the arguments it agent-defends_i, for $i \in \{1, 2\}$.
- $E \subseteq \mathcal{A}$ is an *agent-grounded_i extension* iff it is a minimal-agent complete_i extension (for set inclusion), for $i \in \{1, 2\}$.
- $E \subseteq \mathcal{A}$ is an *agent-preferred_i extension* iff it is a maximal-agent complete_i extension (for set inclusion), for $i \in \{1, 2\}$.
- $E \subseteq \mathcal{A}$ is an *agent-stable_i extension* iff it is conflict-free and it attacks all the arguments in $\mathcal{A} \setminus E$, for $i \in \{1, 2\}$.

The following two examples illustrate agent extensions.

Example 4 (Reinstatement, continued from Example 3). We revisit Figure 2. The individual- and collective-agent complete extension is $\{c\}$. It is also the unique individual- and collective-agent grounded and preferred extension. The individual- and collective-agent stable extension is $\{a, c\}$ ¹.

The following example shows the difference between individual and collective agent defense. In particular, it illustrates that if an argument is defended by a set of arguments all coming from the same agent (individually agent-defended), then the criteria for collective agent defense are automatically satisfied, though the reverse is not always true.

Example 5 (Collective defense). Consider the agent argumentation framework visualized in Figure 3, where $\mathcal{A} = \{a, b_1, b_2, c_1, c_2\}$, where $\rightarrow = \{c_1 \rightarrow b_1, b_1 \rightarrow a, c_2 \rightarrow b_2, b_2 \rightarrow a\}$, where $\mathcal{S} = \{\alpha, \beta, \gamma\}$, and where $\sqsubset = \{(a, \alpha), (a, \beta), (b_1, \gamma), (b_2, \gamma), (c_1, \alpha), (c_2, \beta)\}$. Let us recall the mapping of agents for clarity: α is Alice, β is Bob, and γ is Cayrol. In this variant of the dinner scenario, both Alice and Bob support having meat (argument a), while Cayrol puts forward two arguments: b_1 opposing having meat and b_2 supporting having fish instead. Cayrol's arguments are in turn attacked by both Alice and Bob: Alice attacks b_1 with c_1 , and Bob attacks b_2 with c_2 .

The pair $\{c_1, c_2\}$ *collectively* agent-defends a : they attack all of a 's attackers together. No single agent holding a has all the arguments needed to individually defend it. The agent admissible₁ extensions are $\emptyset$, $\{c_1\}$, $\{c_2\}$, and $\{c_1, c_2\}$. Since a is not individually agent-defended, no admissible₁ set containing a can attack all arguments outside that set, and therefore there is no stable₁ extension. The only agent-complete₁ extension is $\{c_1, c_2\}$, which is also the agent-grounded₁ extension and the unique agent-preferred₁ extension. The agent-admissible₂ extensions are $\emptyset$, $\{c_1\}$, $\{c_2\}$, $\{c_1, c_2\}$, and $\{a, c_1, c_2\}$. The only agent-complete₂ extension is $\{a, c_1, c_2\}$, which is also the grounded₂, preferred₂, and stable₂ extension. Although Alice and Bob do not defend one other's arguments in the sense of individual agent defense, their combined attacks on Cayrol's arguments suffice—under collective agent defense—to reinstate their common argument a .

¹We thank the reviewers for pointing out that in the present framework, the notions of conflict-freeness and attack are defined without reference to agents. As a result, *stable₁* and *stable₂* are the same

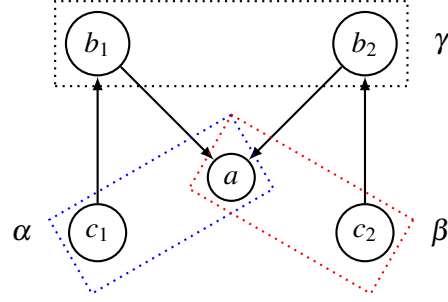


Fig. 3. c_1 and c_2 collectively defend a , but neither can agent-individually defend it.

The following example illustrates another aspect of agent defense.

Example 6 (Individual- and collective-agent defense). Consider Figure 4, where $\mathcal{A} = \{a_1, a_2, b, c\}$, where $\rightarrow = \{c \rightarrow b, b \rightarrow a_2, b \rightarrow a_1\}$, where $\mathcal{S} = \{\alpha, \beta, \gamma\}$ and where $\sqsubseteq = \{(a_1, \alpha), (a_2, \gamma), (b, \beta), (c, \alpha)\}$. The individual-agent complete extension, grounded extension and preferred extension is $\{a_1, c\}$. It is also the unique collective-agent complete extension, grounded extension, because a_2 does not belong to agent α .

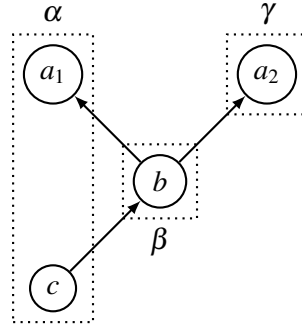


Fig. 4. Agent defense

4. Traditional Agent Argumentation Semantics

In this section, we introduce three existing semantics that extend abstract argumentation with agents. Each is defined via a reduction that transforms an agent argumentation framework into a standard Dung-style argumentation framework, using different ways of incorporating the concept of agent into the reduction process.

4.1. Social Agent Semantics

We begin with *social agent semantics*, which reduces an agent argumentation framework to a preference-based argumentation framework by counting the number of agents supporting each argument. This approach interprets agent argumentation as a form of voting, as studied in social choice

theory and judgment aggregation. While other definitions of social agent semantics are available, the version adopted here is the simplest and the most natural choice for our formal setting. We first recall the definition of preference-based argumentation framework. It is not the only way to define social agent semantics, but given the formal setting we have adopted, it seems the simplest and the most natural possibility.

We first give the definition for a preference-based argumentation framework [29].

Definition 9 ((Preference-based argumentation framework)). A preference-based argumentation framework (PAF) is a 3-tuple $\langle \mathcal{A}, \rightarrow, \succ \rangle$ where $\mathcal{A}$ is a set of arguments, where $\rightarrow \subseteq \mathcal{A} \times \mathcal{A}$ is a binary attack relation, and where $\succ$ is a partial order (irreflexive and transitive) over $\mathcal{A}$ called a preference relation.

Amgoud and Vesic [2] introduced two different reductions of preference, and Van der Torre and Vesic [44] introduced two more. We refer to these articles for explanation and motivation. We illustrate the difference between the reductions in Example 7 below.

Definition 10 (Preference reductions (PR): Reductions of PAF to AF). Given a $PAF = \langle \mathcal{A}, \rightarrow, \succ \rangle$:

- $PR_1(PAF) = \langle \mathcal{A}, \rightarrow' \rangle$, where $\rightarrow' = \{a \rightarrow' b \mid a \rightarrow b, b \not\succ a\}$.
- $PR_2(PAF) = \langle \mathcal{A}, \rightarrow' \rangle$, where $\rightarrow' = \{a \rightarrow' b \mid a \rightarrow b, b \not\succ a \text{ or } b \rightarrow a, \text{ not } a \rightarrow b, a \succ b\}$.
- $PR_3(PAF) = \langle \mathcal{A}, \rightarrow' \rangle$, where $\rightarrow' = \{a \rightarrow' b \mid a \rightarrow b, b \not\succ a \text{ or } a \rightarrow b, \text{ not } b \rightarrow a\}$.
- $PR_4(PAF) = \langle \mathcal{A}, \rightarrow' \rangle$, where $\rightarrow' = \{a \rightarrow' b \mid a \rightarrow b, b \not\succ a, \text{ or } b \rightarrow a, \text{ not } a \rightarrow b, a \succ b, \text{ or } a \rightarrow b, \text{ not } b \rightarrow a\}$.

In social agent semantics, an argument is preferred to another argument if it is held by more agents. The reduction from AAF to PAF is used as an intermediary step for social agent semantics.

Definition 11 (SAP: Social Reductions of AAF to PAF). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubseteq \rangle$, then $SAP(AAF) = \langle \mathcal{A}, \rightarrow, \succ \rangle$ with $\succ = \{a \succ b \mid |\mathcal{S}_a| > |\mathcal{S}_b|\}$.

There are four definitions of social reduction, and σ is in $\{c, g, p, s\}$, thus, we have sixteen social agent semantics.

Definition 12 (SR: Social Reductions of AAF to AF). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubseteq \rangle$, we define $SR_i(AAF) = PR_i(SAP(AAF))$, where PR_i is one of the four reductions of PAF to AF. The semantics is then given by $\delta(AAF) = \sigma(SR_i(AAF)) = \sigma(PR_i(SAP(AAF)))$, for $i \in \{1, 2, 3, 4\}$.

Example 7 (Social reasoning). Consider the agent argumentation framework (AAF) on the left in Figure 5, where $\mathcal{A} = \{a, b\}$, where $\rightarrow = \{a \rightarrow b\}$, where $\mathcal{S} = \{\alpha, \beta\}$ and where $\sqsubseteq = \{(a, \alpha), (b, \alpha), (b, \beta)\}$. Argument b is preferred to argument a because it is held by more agents. The preference-based argumentation framework (PAF) is visualized to the right of the AAF in Figure 5: $\mathcal{A} = \{a, b\}$, and $\rightarrow = \{a \rightarrow b\}$, and $\succ = \{b \succ a\}$. To the right of PAF, there are four argumentation frameworks (AFs) corresponding to SR_1 to SR_4 , and the extensions of each are listed in Table 1.

4.2. Agent Reduction Semantics

Agent-reduction approaches take the perspective of each agent individually, creating a standard argumentation framework for each agent view. Intuitively, each agent prefers their own arguments over those

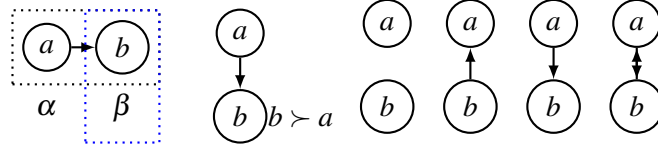


Fig. 5. Social reduction

Sem.	C	G	P	S
SR ₁	$\{\{a, b\}\}$	$\{\{a, b\}\}$	$\{\{a, b\}\}$	$\{\{a, b\}\}$
SR ₂	$\{\{b\}\}$	$\{\{b\}\}$	$\{\{b\}\}$	$\{\{b\}\}$
SR ₃	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$
SR ₄	$\{\emptyset, \{a\}, \{b\}\}$	$\emptyset$	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$

Table 1

The semantics of the four argumentation frameworks (AFs) corresponding to SR_1 to SR_4 . We refer to Dung's semantics as follows: complete (C), grounded (G), preferred (P), stable (S), and the same convention holds for all the others.

of other agents. Then we compute the collective acceptance of each argument. Like social agent semantics, this approach is based on reduction from an agent argumentation framework (AAF) to a standard Dung-style argumentation framework, but it works in a completely different way. In the second volume of the *Handbook of Formal Argumentation*, Baumeister et al. [13] discuss the approaches considered here within a broader context. Essentially, there are two distinct ways to combine the results of these per-agent reductions (cf. Figure 6).

The argument-wise approach determines argument acceptability within individual views using standard methods, and then defines semantic aggregation methods.

The framework-wise approach defines structural aggregation methods to aggregate individual views into a collective representation first, and then determines argument acceptability within the collective representation, using either standard methods or methods created specifically for that representation.

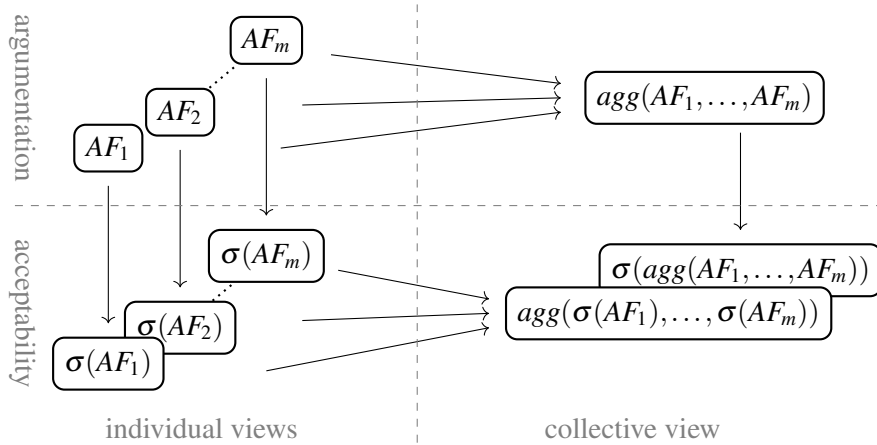


Fig. 6. A schematic overview of different approaches to collective acceptability in formal argumentation, where σ is a general placeholder for some kind of acceptability criterion on AFs, and agg is a general placeholder for some aggregation operator on AFs [13].

Each of the four kinds of reduction of preference-based argumentation frameworks leads to a corresponding kind of agent reduction.

Definition 13 (AAP: Agent Reductions of AAF to PAF). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, we define $AAP(AAF, \alpha) = \langle \mathcal{A}, \rightarrow, \succ \rangle$ with $\succ = \{a \succ b \mid a \sqsubset \alpha \text{ and } \text{not } b \sqsubset \alpha\}$.

As in social agent semantics, there are four definitions of agent reductions, and σ is in $\{c, g, p, s\}$. Thus, we have sixteen agent reduction semantics for each approach.

Definition 14 (AR^F : Framework-Wise Agent Reductions of AAF to AF). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, for $\alpha \in \mathcal{S}$, the object PR_i is one of the four reductions of PAF to AF, where the semantics is defined as $\delta^F(AAF) = \sigma(AR_i^F(AAF)) = \sigma(\bigcup_{\alpha \in \mathcal{S}} PR_i(AAP(AAF, \alpha)))$ for $i \in \{1, 2, 3, 4\}$. For $AF_1 = \langle \mathcal{A}_1, \rightarrow_1 \rangle$ and $AF_2 = \langle \mathcal{A}_2, \rightarrow_2 \rangle$, let $AF_1 \cup AF_2 = \langle \mathcal{A}_1 \cup \mathcal{A}_2, \rightarrow_1 \cup \rightarrow_2 \rangle$.

Example 8 (Framework-Wise Agent reduction). We revisit the AAF on the left in Figure 5. First, we consider the reduction for agent β . Argument b is preferred to argument a ; thus, the resulting PAF is the same as in Figure 5, though for a very different reason than in the case of social reduction. For agent α , the PAF makes all arguments equivalent, and the AF is simply the same as for trivial reduction. To compute the agent extensions of the AAF, we take the union of the reductions for each agent. The AFs of AR_i^F consist of the union of the AFs of SR_i in Table 1 together with the AF in which a attacks b (the reduction for agent α). Thus, $AR_1^F = AR_3^F = \langle \{a, b\}, \{a \rightarrow b\} \rangle$, and $AR_2^F = AR_4^F = \langle \{a, b\}, \{a \rightarrow b, b \rightarrow a\} \rangle$. For instance, after AR_1^F , the AF of agent α is $AR_1^F(AAF, \alpha) = \langle \{a, b\}, \{a \rightarrow b\} \rangle$, and the AF of agent β is $AR_1^F(AAF, \beta) = \langle \{a, b\}, \{\emptyset\} \rangle$. So the union is $\langle \{a, b\}, \{a \rightarrow b\} \rangle$, from which we then compute the extensions of this union. The result is Table 2 below for the sixteen agent reduction semantics under consideration.

Sem.	C	G	P	S
AR_1^F	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$
AR_2^F	$\{\emptyset, \{a\}, \{b\}\}$	$\{\emptyset\}$	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$
AR_3^F	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$
AR_4^F	$\{\emptyset, \{a\}, \{b\}\}$	$\{\emptyset\}$	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$

Table 2

Framework-wise complete (C), grounded (G), preferred (P) and stable (S) semantics of four argumentation frameworks (AF) corresponding to AR_1^F to AR_4^F .

The argument-wise approach combines agent reductions in a new way. Instead of combining the frameworks, we can take the union of all the extensions to the individual frameworks:

Definition 15 (AR^A : Argument-Wise Agent Reductions of AAF to AF). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, for $\alpha \in \mathcal{S}$, PR_i is one of the four reductions of PAF to AF, where the semantics $\delta^A(AAF) = \sigma(AR_i^A(AAF)) = \bigcup_{\alpha \in \mathcal{S}} \sigma(PR_i(AAP(AAF, \alpha)))$ for $i \in \{1, 2, 3, 4\}$.

If we use an argument-wise approach to the AAF on the left in Figure 5, the results as shown in Table 3 are different to those of Table 2.

Sem.	C	G	P	S
AR_1^A	$\{\{a\}, \{a, b\}\}$	$\{\{a\}, \{a, b\}\}$	$\{\{a\}, \{a, b\}\}$	$\{\{a\}, \{a, b\}\}$
AR_2^A	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$
AR_3^A	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$	$\{\{a\}\}$
AR_4^A	$\{\emptyset, \{a\}, \{b\}\}$	$\{\{a\}, \emptyset\}$	$\{\{a\}, \{b\}\}$	$\{\{a\}, \{b\}\}$

Table 3

Argument-Wise complete (C), grounded (G), preferred (P) and stable (S) semantics of four argumentation frameworks (AF) corresponding to AR_1^A to AR_4^A .

4.3. Agent Filtering Semantics

In this section, we introduce the fourth kind of semantics for agent argumentation frameworks. Intuitively, agents may hold or express incomplete or even inconsistent information, especially in complex, uncertain and dynamic environments. After all, formal argumentation originates from conflict, and its primary aim is to represent and manage reasoning inconsistencies and disagreements regardless of whether they occur at the individual or collective level.

Agent filtering semantics addresses this by removing arguments or attacks that are not attributed to any agent, and are therefore considered either unknown or not trusted. This interpretation is particularly useful in design-time scenarios where the aim is to assess the internal coherence of individual agents' arguments before considering their interactions with others. In such settings, cross-agent attacks may be irrelevant because they are never considered together by a single decision-maker.

The OrphanReduction function removes *arguments* that are not held by any agent. The NotBothReduction function removes *attacks* that are not held by any agent. An attack is said to belong to an agent only if the target and attacker arguments are both held by the same agent. The reasoning behind this is that if two compatible arguments are held by different agents and these agents are not engaged in a dialogue and do not have knowledge of one other's arguments, the existence of the attack is simply unknown from the perspective of any individual agent. In other words, attacks across agents require interaction or shared awareness, and in their absence, it is reasonable to omit such attacks from the epistemic standpoint of each agent.

Definition 16 (Agent Reductions of AAF to AF). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$:

- OrphanRemoval (OR): $OR(AAF) = \langle \mathcal{A}', \rightarrow' \rangle$, where $\mathcal{A}' = \{a \mid \exists \alpha \in \mathcal{S} \text{ such that } a \sqsubset \alpha\}$, $\rightarrow' \cap \mathcal{A}' \times \mathcal{A}'$.
- NotBothReduction (NBR): $NBR(AAF) = \langle \mathcal{A}, \rightarrow' \rangle$, where $\rightarrow' = \{(a \rightarrow b) \mid \exists \alpha \in \mathcal{S} \text{ such that } a \sqsubset \alpha, \text{ and } b \sqsubset \alpha\}$.

Example 9 (Epistemic reasoning). Consider the two AAFs in Figure 7. For the figure on the left, we may say that argument a is unknown because it is not held by any agent, and for the figure on the right, we may say that the attack is unknown because there is no agent holding both arguments a and b . The filtering methods remove such unknown arguments (OrphanReduction) and unknown attacks (NotBothReduction).

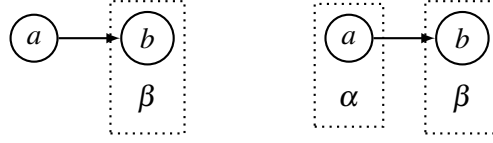


Fig. 7. Unknown argument and unknown attack

5. Traditional Principles

In this section, we repeat six important principles from the literature. As the baseline for the principles, we also include Dung's semantics based on trivial reduction, i.e., we simply ignore the agents and the relation between agents and arguments.

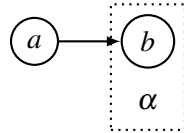
Definition 17 (TR: Trivial Reduction). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, then $TR(AAF) = \langle \mathcal{A}, \rightarrow \rangle$.

Principle 1 (Conflict-free [10]). An agent semantics δ satisfies the conflict-free principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, for all $E \in \delta(AAF)$, there are no arguments a and b in E such that a attacks b .

The conflict-free principle reflects the intuitive idea that an extension contains the arguments that can be accepted together and that conflicting arguments cannot be included in the same extension. The admissibility principle reflects the idea that all arguments are defended.

Proposition 1. None of the four kinds of Dung semantics for SR_1 (AR_1^F , AR_1^A , NBR) satisfies the conflict-free principle.

Proof. Consider the AAF in Figure 8. It is easy to see that $SR_1(AAF) = PR_i(AAP(AAF, \alpha)) = NBR(AAF) = \langle \mathcal{A}, \emptyset \rangle$. Note that the complete, grounded, preferred, and stable extensions of $\langle \mathcal{A}, \emptyset \rangle$ are all $\{a, b\}$, but they are not conflict-free in AAF . $\square$

Fig. 8. An $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$.

Principle 2 (Admissibility [10]). An agent semantics δ satisfies the admissibility principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, every $E \in \delta(AAF)$ is admissible in $\langle \mathcal{A}, \rightarrow \rangle$.

It follows from Proposition 1 that none of the four kinds of Dung semantics for SR_1 (AR_1^F , AR_1^A , NBR) satisfies the admissibility principle. Moreover:

Proposition 2. None of the four kinds of Dung semantics for SR_2 (AR_2^F , AR_2^A , OR) satisfies the admissibility principle.

Proof. We also consider the AAF in Figure 8. Here, $SR_2(AAF) = PR_i(AAP(AAF, \alpha)) = \langle \mathcal{A}, \{(b, a)\} \rangle$. Note that $\{b\}$ comprises the complete, grounded, preferred, and stable extensions of $\langle \mathcal{A}, \{(b, a)\} \rangle$. However, $\{b\}$ is not admissible in $\langle \mathcal{A}, \rightarrow \rangle$.

Besides, since $OR(AAF) = \langle \{b\}, \emptyset \rangle$, then $\{b\}$ comprises the complete, grounded, preferred, and stable extensions of $OR(AAF)$. But $\{b\}$ is not admissible in $\langle \mathcal{A}, \rightarrow \rangle$. $\square$

Directionality and SCC-recursiveness were introduced by Baroni, Giacomin, and Guida [11]. These principles reflect the idea that we can decompose an argumentation framework into sub-frameworks so that the semantics can be defined locally. For the directionality principle, they first introduced the definition of an unattacked set.

Definition 18 (Unattacked Set). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, a set $\mathcal{U}$ is unattacked iff there exists no $a \in \mathcal{A} \setminus \mathcal{U}$ such that a attacks an argument in $\mathcal{U}$. The set of unattacked sets in AAF is denoted as $\mathcal{US}(AAF)$.

Definition 19 (Restriction). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, and letting $\mathcal{U} \subseteq \mathcal{A}$ be a set of arguments, the restriction of AAF to $\mathcal{U}$ is the agent argumentation framework $AAF \downarrow_{\mathcal{U}} = \langle \mathcal{U}, \rightarrow \cap \mathcal{U} \times \mathcal{U}, \mathcal{S} \cap \mathcal{U} \times \mathcal{U} \rangle$.

Principle 3 (Directionality [10]). An agent semantics δ satisfies the directionality principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, for every $\mathcal{U} \in \mathcal{US}(AAF)$, it holds that $\delta(AAF \downarrow_{\mathcal{U}}) = \{E \cap \mathcal{U} \mid E \in \delta(AAF)\}$.

Proposition 3. Agent $stable_1$ semantics and agent $stable_2$ semantics (Def. 7) do not satisfy Principle 3.

Proof. We propose a counterexample. Assume an $AAF = \langle \{a_1, a_2, a_3, b\}, \{b \rightarrow a_3, a_3 \rightarrow a_1, a_1 \rightarrow a_2, a_2 \rightarrow a_3\}, \{\alpha\}, \{b \sqsubset \alpha, a_1 \sqsubset \alpha, a_2 \sqsubset \alpha, a_3 \sqsubset \alpha\} \rangle$ (see Figure 5). The unattacked set of arguments is $\mathcal{U} = \{b\}$. The stable extension of $(AAF \downarrow_{\mathcal{U}})$ is $\{b\}$. However, there is no stable extension of this AAF . Because $\delta(AAF \downarrow_{\mathcal{U}}) \neq \{E \cap \mathcal{U} \mid E \in \delta(AAF)\}$, agent $stable_1$ semantics and agent $stable_2$ semantics do not satisfy Principle 3. $\square$

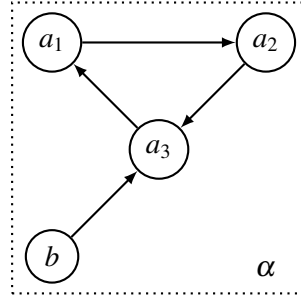


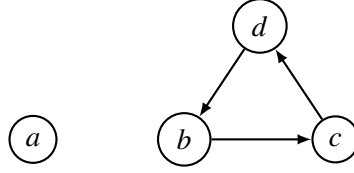
Fig. 9. An AAF.

Proposition 4 ($SR_1, SR_3 \times P_3$). The grounded, complete, preferred semantics for SR_1 and SR_3 satisfy Principle 3 but the stable semantics for SR_1 and SR_3 do not satisfy Principle 3.

Proof. For any $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, if a set of arguments $\mathcal{U} \subseteq \mathcal{A}$ is unattacked in AAF , then it is also unattacked in both $SR_1(AAF)$ and $SR_3(AAF)$. Thus, the first half follows from [10] directly.

For the second half of the lemma, consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 10. It is easy to see that $SR_1(AAF) = SR_3(AAF) = \langle \mathcal{A}, \rightarrow \rangle$. The set $\{a\}$ is unattacked and is a stable extension of $AAF \downarrow_{\{a\}}$. But there is no stable extension E of $\langle \mathcal{A}, \rightarrow \rangle$ such that $a \in E$. $\square$

SCC-recursiveness is based on the concept of strongly connected components from graph theory.

Fig. 10. Stable semantics for SR_1 and SR_3 do not satisfy Principle 3.

Definition 20 (Strongly Connected Component). Let an AAF be $\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$. The binary relation of path-equivalence between nodes, denoted as $PE_{AAF} \subseteq (\mathcal{A} \times \mathcal{A})$, is defined as follows:

- for every $a \in \mathcal{A}$, it is the case that $(a, a) \in PE_{AAF}$
- given two distinct arguments $a, b \in \mathcal{A}$, we say that $(a, b) \in PE_{AAF}$ iff there is a path from a to b and a path from b to a .

The strongly connected components of AAF are the equivalence classes of arguments under the relation of path-equivalence. The set of strongly connected components is denoted by $SCCS_{AAF}$.

Given an argument $a \in \mathcal{A}$, the notation $SCCS_{AAF}(a)$ stands for the strongly connected component that contains a . In the particular case where the argumentation framework is empty, i.e., $AAF = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$, we assume that $SCCS_{AAF} = \{\emptyset\}$. The choice of extensions of the antecedent's strongly connected components determines a partition of the arguments of a strongly connected component S into three subsets: defeated (D), provisionally defeated (P) and undefeated (U) [11].

In other words, the set $D_{AAF}(S, E)$ consists of the arguments of S being attacked by E from outside S . The set $U_{AAF}(S, E)$ consists of the arguments in S that are not attacked by E from outside S and that are defended by E . And $P_{AAF}(S, E)$ consists of the arguments in S that are not attacked by E from outside S and that are not defended by E .

Definition 21 (D, P, U, UP). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, a set $E \subseteq \mathcal{A}$, and a strongly connected component $S \in SCCS_{AAF}$:

- $D_{AAF}(S, E) = \{a \in S \mid (E \cap S_{out}^-) \text{ attacks } a\}$
- $P_{AAF}(S, E) = \{a \in S \mid (E \cap S_{out}^-) \text{ does not attack } a \text{ and } \exists b \in (S_{out}^- \cap a^-) \text{ such that } E \text{ does not attack } b\}$.
- $U_{AAF}(S, E) = S \setminus (D_{AAF}(S, E) \cup P_{AAF}(S, E))$
- $UP_{AAF}(S, E) = U_{AAF}(S, E) \cup P_{AAF}(S, E)$.

We now present the notion of SCC-recursiveness, which was introduced by Baroni, Giacomin, and Guida [11].

Principle 4 (SCC-recursiveness [11]). Agent semantics δ satisfies the SCC-recursiveness principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, we have $\delta(AAF) = \mathcal{G}(AAF, \mathcal{A})$ where, for every AAF and for every set $\mathcal{C} \subseteq \mathcal{A}$, the function $\mathcal{G}(AAF, \mathcal{C}) \subseteq 2^{\mathcal{A}}$ is defined as follows. For every $E \subseteq \mathcal{A}$, it holds that $E \in \mathcal{G}(AAF, \mathcal{C})$ iff

- when $|SCCS_{AAF}| = 1$, we have that $E \in \mathcal{B}(AAF, \mathcal{C})$,
- otherwise, $\forall S \in SCCS_{AAF}, (E \cap S) \in \mathcal{G}(AAF \downarrow_{UP_{AAF}(S, E)}, U_{AAF}(S, E) \cap \mathcal{C})$,

where $\mathcal{B}(AAF, \mathcal{C})$ is a function called a base function that, given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ such that $|SCCS_{AAF}| = 1$ and a set $\mathcal{C} \subseteq \mathcal{A}$, gives a subset of $2^{\mathcal{A}}$.

Proposition 5 ($\text{Sem}_1, \text{Sem}_2 \times P_4$). The grounded, complete and preferred semantics under admissibility₁ and/or admissibility₂ do not satisfy Principle 4, but the stable semantics does satisfy Principle 4.

Proof. Since the stable semantics is the same as in abstract argumentation frameworks, the second half follows from [11] directly. Specifically, we can define the function $\mathcal{G}$ in such a way that $\mathcal{G}(\text{AAF}, \mathcal{C}) = SE(\text{TR}(\text{AAF}), \mathcal{C})$, where the function SE is defined on page 184 of [11]. Consider $\text{AAF} = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 11. It is clear that, under both admissibility₁ and admissibility₂, $\{a\}$ is the grounded extension

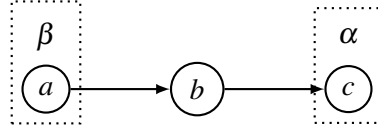


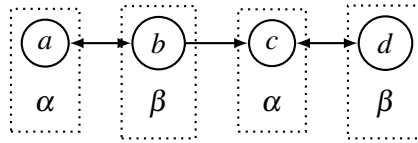
Fig. 11. An AAF

of the above AAF, as well as its complete and preferred extension. We first consider the grounded semantics. Suppose, toward a contradiction, that there is a function $\mathcal{G}$ as described in Principle 4. Thus, $\mathcal{G}(\text{AAF}, \mathcal{A}) = \{\{a\}\}$ (1). On the other hand, let $E \in \mathcal{G}(\text{AAF}, \mathcal{A})$. There are three SCCs in the AAF, i.e., $S_1 = \{a\}$, and $S_2 = \{b\}$, and $S_3 = \{c\}$. By (1), we know that $E \cap S_1 = \{a\}$. Now consider the SCC S_2 . Since $UP_{\text{AAF}}(S_2, E) = U_{\text{AAF}}(S_2, E) = \emptyset$, then $E \cap S_2 = \emptyset$. And S_3 remains to be considered. Note that $UP_{\text{AAF}}(S_3, E) = U_{\text{AAF}}(S_3, E) = \{c\}$. So $E \cap S_3 \in \mathcal{G}(\text{AAF} \downarrow_{\{c\}}, \{c\})$. Note that $\mathcal{G}(\text{AAF} \downarrow_{\{c\}}, \{c\})$ must be the grounded extension of $\text{AAF} \downarrow_{\{c\}}$, which is $\{c\}$. Thus, $E = \{a, c\}$, which contradicts (1).

The cases for the complete and preferred semantics can be shown in a similar way. $\square$

Proposition 6 ($AR_3^A \times P_4$). None of the four kinds of Dung semantics for AR_3^A satisfies Principle 4.

Proof. We first consider the grounded semantics. Suppose, toward a contradiction, that the grounded semantics for AR_3^A satisfies Principle 4. Then there exists a function $\mathcal{G}$ as described in Principle 4. Consider the following AAF $= \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$:



We have:

- (1) $\mathcal{G}(\text{AAF}, \mathcal{A}) = \{B \subseteq \mathcal{A} \mid \exists \alpha \in \mathcal{S} : B \text{ is a grounded extension of } PR_3(\text{AAP}(\text{AAF}, \alpha))\} = \{\{a, c\}, \{b, d\}\}.$
- (2) $\mathcal{G}(\text{AAF} \downarrow_{\{a, b\}}, \{a, b\}) = \{B \subseteq \{a, b\} \mid \exists \alpha \in \mathcal{S} : B \text{ is a grounded extension of } PR_3(\text{AAP}(\text{AAF} \downarrow_{\{a, b\}}, \alpha))\} = \{\{a\}, \{b\}\}.$
- (3) $\mathcal{G}(\text{AAF} \downarrow_{\{c, d\}}, \{c, d\}) = \{B \subseteq \{c, d\} \mid \exists \alpha \in \mathcal{S} : B \text{ is a grounded extension of } PR_3(\text{AAP}(\text{AAF} \downarrow_{\{c, d\}}, \alpha))\} = \{\{c\}, \{d\}\}.$

Now consider the set $E = \{a, d\}$. We show that:

$$\forall S \in \text{SCCS}_{\text{AAF}}, (E \cap S) \in \mathcal{G}(\text{AAF} \downarrow_{UP_{\text{AAF}}(S, E)}, U_{\text{AAF}}(S, E) \cap \mathcal{A}).$$

If $S = \{a, b\}$, then $UP_{AAF}(S, E) = U_{AAF}(S, E) = \{a, b\}$. Hence $(E \cap S) \in \mathcal{G}(AAF \downarrow_{UP_{AAF}(S, E)}, U_{AAF}(S, E) \cap \mathcal{A})$. Otherwise, $S = \{c, d\}$. We have $UP_{AAF}(S, E) = U_{AAF}(S, E) = \{c, d\}$. Therefore, we also have $(E \cap S) \in \mathcal{G}(AAF \downarrow_{UP_{AAF}(S, E)}, U_{AAF}(S, E) \cap \mathcal{A})$. However, note that $E \notin \mathcal{G}(AAF, \mathcal{A})$. A contradiction!

The cases for complete, preferred, and stable semantics can be shown in a similar way. $\square$

Baumann, Brewka, and Ulbricht [12] introduce the modularization principle. By definition, AAF^E is the sub-framework of AAF obtained by removing the so-called range of E, the corresponding attacks, and the relation with agents.

Definition 22 (E-reduct). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and $E \subseteq \mathcal{A}$, let $E^+ = \{a \in \mathcal{A} \mid E \text{ attacks } a\}$, let $E^\oplus = E \cup E^+$, and let $E^* = \mathcal{A} \setminus E^\oplus$. The E-reduct of AAF is $AAF^E = \langle E^*, R \cap (E^* \times E^*), \mathcal{S}, \sqsubset \cap (\mathcal{S} \times E^*) \rangle$.

Principle 5 (Modularity). An agent semantics δ satisfies modularization if for any AAF, it is the case that $E \in \delta(AAF)$ and $E' \in \delta(AAF^E)$ together imply $E \cup E' \in \delta(AAF)$.

The modularity principle is related to the robustness principles of Rienstra et al. [42], which consider the addition and removal of arguments and attacks. We consider here only argument removal, which we call argument modularity.

Proposition 7 ($SR_1 \times P_5$). The grounded, complete, and preferred semantics for SR_1 do not satisfy Principle 5, but the stable semantics for SR_1 does satisfy Principle 5.

Proof. Consider the following $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$:

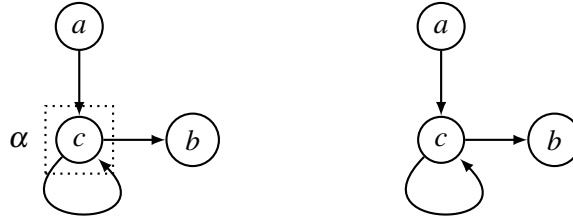


Fig. 12. SR_1 does not satisfy Principle 5.

It is easy to see that $\{a\}$ is the grounded extension of $SR_1(AAF)$ and is also a complete extension and a preferred extension. Let $E = \{a\}$, then AAF^E consists of the single point b . It is also easy to see that $\{b\}$ constitutes the grounded, complete and preferred extensions of AAF^E . However, $\{a, b\}$ is not admissible in $SR_2(AAF)$.

For the second half of the proposition, let $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and let E be a stable extension of $SR_1(AAF)$. It is easy to see that E^* is empty, thus E' is also empty. $\square$

Proposition 8 ($AR_4^A \times P_5$). None of the four kinds of Dung semantics for AR_4^A satisfies Principle 5.

Proof. For the complete, preferred, and stable semantics, we use the same counterexample as in Prop. 43. For the grounded semantics, consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 13. It can be seen that $E = \{b\}$ is the grounded extension of $PR_4(AAP(AAF, \alpha))$ and that $E' = \{c\}$ is the grounded extension of $PR_4(AAP(AAF^E, \beta))$. However, $E \cup E' = \{b, c\}$ is the grounded extension of neither $PR_4(AAP(AAF, \alpha))$ nor $PR_4(AAP(AAF, \beta))$. $\square$

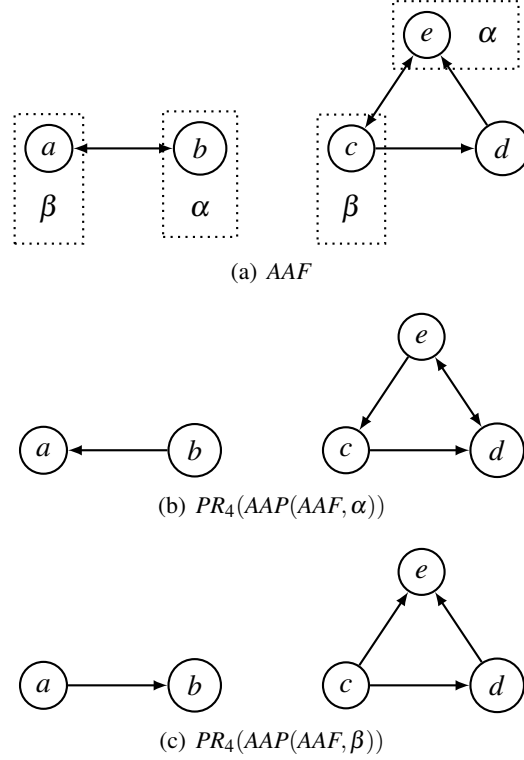


Fig. 13. A counterexample for Prop. 8.

Table 4 provides a full analysis of the traditional five principles. The first line of the trivial reduction presents a well-known analysis of which of these principles hold for Dung's semantics. Unsurprisingly, several easy examples we have already discussed in this article show that few of the traditional principles hold for agent semantics. This is a problem, particularly for SCC-recursiveness and modularity, because we cannot apply the corresponding recursive algorithm to compute the semantics. In the next section, we therefore introduce variants of admissibility, SCC-recursion, and modularity based on agent defense.

6. Variants of Traditional Principles

The agent admissibility principle is a straightforward adaptation of the traditional admissibility principle in which the traditional defense is replaced by agent defense. Since there are two kinds of admissibility, one for individual defense and one for collective defense, we end up with two agent admissibility principles.

Principle 6 (Agent Admissibility₁). An agent semantics δ satisfies the agent admissibility₁ principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, every $E \in \delta(AAF)$ is agent admissible₁.

Principle 7 (Agent Admissibility₂). An agent semantics δ satisfies the agent admissibility₂ principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, every $E \in \delta(AAF)$ is agent admissible₂.

Proposition 9. AR_1^F to AR_4^F and SR_1 to SR_4 do not satisfy Principles 6 and 7 for complete semantics.

Sem.	P1	P2	P3	P4	P5
TR	CGPS	CGPS	CGP	CGPS	CGPS
Sem ₁	CGPS	CGPS	CGP	S	S
Sem ₂	CGPS	CGPS	CGP	S	S
SR ₁	×	×	CGP	×	×
SR ₂	CGPS	×	×	×	×
SR ₃	CGPS	CGPS	CGP	CGPS	CGPS
SR ₄	CGPS	G	×	×	G
AR ₁ ^F	×	×	CGP	×	S
AR ₂ ^F	CGPS	×	×	×	×
AR ₃ ^F	CGPS	CGPS	CGP	CGPS	CGPS
AR ₄ ^F	CGPS	G	×	×	G
AR ₁ ^A	×	×	CGP	×	S
AR ₂ ^A	CGPS	×	×	×	×
AR ₃ ^A	CGPS	CGPS	CGP	×	S
AR ₄ ^A	CGPS	G	×	×	×
OR	CGPS	×	CGP	CGPS	CGPS
NBR	×	×	CGP	×	S

Table 4

A comparison of reductions and traditional principles. When a principle is never satisfied by a certain reduction for all semantics, we use the $\times$ symbol. P1 refers to Principle 1, and the same convention holds for all the others.

Proof. We use the agent argumentation framework in Figure 2 as a counterexample. $SAP(AAF) = AAP(AAF) = \langle \{a, b, c\}, \{a \rightarrow b, b \rightarrow c\}, \emptyset \rangle$. And $AR_i^F(AAF) = SR_i(AAF) = \langle \{a, b, c\}, \{a \rightarrow b, b \rightarrow c\} \rangle$. The complete extension of $AR_i^F(AAF)$ and $SR_i(AAF)$ is $\{a, c\}$. However, a cannot agent-defend c , and $\{a, c\}$ is not agent-admissible. Thus, AR_1^F to AR_4^F and SR_1 to SR_4 do not satisfy Principles 6 and 7 for complete semantics. $\square$

Proposition 10 (Sem₁, Sem₂ $\times$ P₆, P₇). Agent stable₁₋₂ semantics do not satisfy Principle 6 or Principle 7.

Proof. Consider the AAF in Figure 14. It is easy to see that $\{a, b, e\}$ is a stable extension of the AAF.

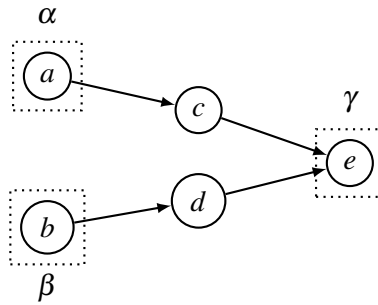


Fig. 14. Agent defense

However, $\{a, b, e\}$ is neither admissible₁ nor admissible₂. $\square$

The agent SCC-recursiveness principles are also adapted by replacing traditional defense with agent defense, and again we end up with two principles for individual and collective defense. What needs to be adapted is the definition of P, the provisionally defeated arguments. Roughly, P stands for cases where an argument is not defended against b in E outside of S . Likewise, AP stands for cases where an argument a is not agent-defended _{i} against b in E from outside S .

To define agent SCC-recursiveness, we define AD_i , AP_i , AU_i and AUP_i under individual agent defense and collective agent defense.

Definition 23 (AD_i , AP_i , AU_i , AUP_i). Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, a set $E \subseteq \mathcal{A}$ and a strongly connected component $S \in SCCS_{AAF}$, we define:

- $AD_{iAAF}(S, E) = D_{iAAF}(S, E)$
- $AP_{1AAF}(S, E) = \{a \in S \mid (E \cap S_{out}^-) \text{ does not attack } a, \text{ and } \forall \alpha \in \mathcal{S}_a, \exists b \in (S_{out}^- \cap a^-) \text{ such that } E \cap \mathcal{A}_{\mathcal{S}_a} \text{ does not attack } b.\}$
- $AP_{2AAF}(S, E) = \{a \in S \mid (E \cap S_{out}^-) \text{ does not attack } a \text{ and } \exists b \in (S_{out}^- \cap a^-) \text{ such that } \forall \alpha \text{ in } \mathcal{S}_a, E \cap \mathcal{A}_\alpha \text{ does not attack } b.\}$
- $AU_{iAAF}(S, E) = S \setminus (AD_{iAAF}(S, E) \cup AP_{iAAF}(S, E))$
- $AUP_{iAAF}(S, E) = AU_{iAAF}(S, E) \cup AP_{iAAF}(S, E)$

Principle 8 (Agent SCC-recursiveness₁). An agent semantics δ satisfies the agent SCC-recursiveness₁ principle iff, for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, we have $\delta(AAF) = \mathcal{G}(AAF, \mathcal{A})$ where, for every AAF and every set $\mathcal{C} \subseteq \mathcal{A}$, the function $\mathcal{G}(AAF, \mathcal{C}) \subseteq 2^{\mathcal{A}}$ is defined as follows. For every $E \subseteq \mathcal{A}$, $E \in \mathcal{G}(AAF, \mathcal{C})$ iff

- when $|SCCS_{AAF}| = 1$, it holds that $E \in \mathcal{B}(AAF, \mathcal{C})$,
- otherwise, $\forall S \in SCCS_{AAF}$, $(E \cap S) \in \mathcal{G}(AAF \downarrow_{AUP_{1AAF}(S, E)}, AU_{1AAF}(S, E) \cap \mathcal{C})$,

Proposition 11 ($Sem_1 \times P_8$). The grounded, complete and preferred semantics under admissibility₁ do not satisfy Principle 8.

Proof. For the complete and preferred semantics, consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 15. We

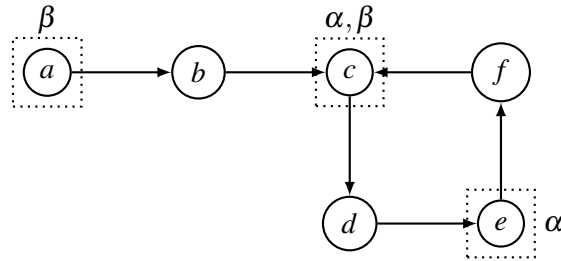


Fig. 15. An AAF

first consider the complete semantics. Suppose, toward a contradiction, that there is function $\mathcal{G}$ as described in Principle 8. Let $E \in \mathcal{G}(AAF, \mathcal{A})$. There are three SCCs: $S_1 = \{a\}$, and $S_2 = \{b\}$, and $S_3 = \{c, d, e, f\}$. For S_1 , since $AUP_{1AAF}^1(S_1, E) = \{a\}$, then $AAF \downarrow_{AUP_{1AAF}^1(S_1, E)}$ consists of the single point a . The complete extension of $AAF \downarrow_{AUP_{1AAF}^1(S_1, E)}$ under admissibility₁ is $\{a\}$. Since $AUP_{1AAF}^1(S_1, E) = AU_{1AAF}^1(S_1, E) = \{a\}$, by Principle 8, it holds that $(E \cap S_1) = \mathcal{G}(AAF \downarrow_{\{a\}}, \{a\}) = \{a\}$. For S_2 , since $AUP_{1AAF}^1(S_2, E) = AU_{1AAF}^1(S_2, E) = \emptyset$, then $(E \cap S_2) = \emptyset$. For S_3 , we first note that $AUP_{1AAF}^1(S_3, E) =$

$AU_{AAF}^1(S_3, E) = \{c, d, e, f\}$. By Principle 8, $\mathcal{G}(AAF \downarrow_{\{c, d, e, f\}}, \{c, d, e, f\})$ must be the set of complete extensions of $AAF \downarrow_{\{c, d, e, f\}}$. Since $\{c, e\}$ is the only complete extension of $AAF \downarrow_{\{c, d, e, f\}}$, we have $E \cap S_3 = \mathcal{G}(AAF \downarrow_{\{c, d, e, f\}}, \{c, d, e, f\}) = \{c, e\}$. In sum, $E = \{a, c, e\}$. However, E cannot defend₁ itself in AAF (consider argument c), a contradiction!

The case for preferred semantics can be shown in a similar way.

For the grounded semantics, consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 16. Suppose, toward a contra-

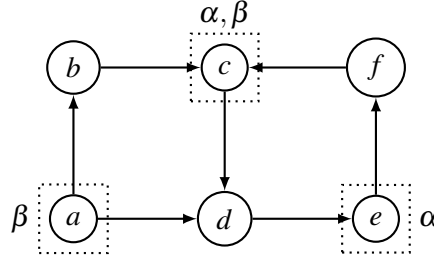


Fig. 16. An AAF

diction, that there is function $\mathcal{G}$ as described in Principle 8. Let $E \in \mathcal{G}(AAF, \mathcal{A})$. There are three SCCs: $S_1 = \{a\}$, and $S_2 = \{b\}$, and $S_3 = \{c, d, e, f\}$. For S_1 , since $AUP_{AAF}^1(S_1, E) = \{a\}$, then $AAF \downarrow_{AUP_{AAF}^1(S_1, E)}$ consists of the single point a . The grounded extension of $AAF \downarrow_{\{a\}}$ under admissibility₁ is $\{a\}$. Since $AUP_{AAF}^1(S_1, E) = AU_{AAF}^1(S_1, E) = \{a\}$, by Principle 8, $(E \cap S_1) = \mathcal{G}(AAF \downarrow_{\{a\}}, \{a\}) = \{a\}$. For S_2 , since $AUP_{AAF}^1(S_2, E) = AU_{AAF}^1(S_2, E) = \emptyset$, then $(E \cap S_2) = \emptyset$. For S_3 , we first note that $AUP_{AAF}^1(S_3, E) = AU_{AAF}^1(S_3, E) = \{c, e, f\}$. By Principle 8, $\mathcal{G}(AAF \downarrow_{\{c, e, f\}}, \{c, e, f\})$ must be the grounded extension of $AAF \downarrow_{\{c, e, f\}}$. Since the grounded extension of $AAF \downarrow_{\{c, e, f\}}$ is $\{c, e\}$, we have $E \cap S_3 = \mathcal{G}(AAF \downarrow_{\{c, e, f\}}, \{c, e, f\}) = \{c, e\}$. In sum, $E = \{a, c, e\}$. However, E cannot defend₁ itself in the AAF (consider argument c), a contradiction! $\square$

Principle 9 (Agent SCC-recursiveness₂). An agent semantics δ satisfies the agent SCC-recursiveness₂ principle iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, we have $\delta(AAF) = \mathcal{G}(AAF, \mathcal{A})$, where for every AAF and for every set $\mathcal{C} \subseteq \mathcal{A}$, the function $\mathcal{G}(AAF, \mathcal{C}) \subseteq 2^{\mathcal{A}}$ is defined as follows. For every $E \subseteq \mathcal{A}$, then $E \in \mathcal{G}(AAF, \mathcal{C})$ iff:

- when $|SCCS_{AAF}| = 1$, it holds that $E \in \mathcal{B}(AAF, \mathcal{C})$;
- otherwise, $\forall S \in SCCS_{AAF}$, $(E \cap S) \in \mathcal{G}(AAF \downarrow_{AUP_{2AAF}(S, E)}, AU_{2AAF}(S, E) \cap \mathcal{C})$.

Proposition 12 ($\text{Sem}_2 \times \text{P}_9$). The complete, grounded and preferred semantics for admissibility₂ do not satisfy Principle 9.

Proof. For the grounded semantics, consider $\text{AAF} = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 16. Suppose, toward a contradiction, that there is function $\mathcal{G}$ as described in Principle 9. Let $E \in \mathcal{G}(\text{AAF}, \mathcal{A})$. There are three SCCs: $S_1 = \{a\}$, and $S_2 = \{b\}$, and $S_3 = \{c, d, e, f\}$. For S_1 , since $\text{AUP}_{\text{AAF}}^2(S_1, E) = \{a\}$, then $\text{AAF} \downarrow_{\text{AUP}_{\text{AAF}}^2(S_1, E)}$ consists of the single point a . The grounded extension of $\text{AAF} \downarrow_{\{a\}}$ under admissibility₂ is $\{a\}$. Since $\text{AUP}_{\text{AAF}}^2(S_1, E) = \text{AU}_{\text{AAF}}^2(S_1, E) = \{a\}$, by Principle 9, $(E \cap S_1) = \mathcal{G}(\text{AAF} \downarrow_{\{a\}}, \{a\}) = \{a\}$. For S_2 , since $\text{AUP}_{\text{AAF}}^2(S_2, E) = \text{AU}_{\text{AAF}}^2(S_2, E) = \emptyset$, then $(E \cap S_2) = \emptyset$. For S_3 , we first note that $\text{AUP}_{\text{AAF}}^2(S_3, E) = \text{AU}_{\text{AAF}}^2(S_3, E) = \{c, e, f\}$. By Principle 8, $\mathcal{G}(\text{AAF} \downarrow_{\{c, e, f\}}, \{c, e, f\})$ must be the grounded extension of $\text{AAF} \downarrow_{\{c, e, f\}}$. Since the grounded extension of $\text{AAF} \downarrow_{\{c, e, f\}}$ (under admissibility₂) is $\{c, e\}$, we have $E \cap S_3 = \mathcal{G}(\text{AAF} \downarrow_{\{c, e, f\}}, \{c, e, f\}) = \{c, e\}$. In sum, $E = \{a, c, e\}$. However, E cannot defend₂ itself in AAF (consider argument d), a contradiction!

Similar arguments hold for the complete and preferred semantics. $\square$

Table 5 shows the comparison between agent semantics, agent admissibility principles, and agent SCC-recursion. This is important because it proves that we can have an efficient SCC-recursiveness algorithm for the new agent semantics. The table also shows that for P7 and P9, collective defense implies individual defense. Moreover, it shows that the adapted principles, like the traditional ones, are not very useful for distinguishing between the reduction-based semantics, i.e., the social agent semantics, the agent reduction semantics, and the agent filtering semantics. Therefore, we introduce some new principles in the remainder of the article.

Sem.	P6	P7	P8	P9
TR	×	×	×	×
Sem ₁	CGP	CGP	\$	\$
Sem ₂	×	CGP	\$	\$
SR ₁	×	×	×	×
SR ₂	×	×	×	×
SR ₃	×	×	×	×
SR ₄	×	×	×	×
AR ₁ ^F	×	×	×	×
AR ₂ ^F	×	×	×	×
AR ₃ ^F	×	×	×	×
AR ₄ ^F	×	×	×	×
AR ₁ ^A	×	×	×	×
AR ₂ ^A	×	×	×	×
AR ₃ ^A	×	×	×	×
AR ₄ ^A	×	×	×	×
OR	×	×	×	×
NBR	×	×	×	×

Table 5

Comparison of reductions, agent admissibility principles (P6–P9), and agent SCC-recursion.

Sem.	P10	P11	P12	P13	P14	P15	P16	P17
TR	CGPS	CGPS	CGPS	CGPS	CGPS	CGPS	×	×
Sem ₁	S	S	CGPS	×	CGPS	×	×	×
Sem ₂	S	S	CGPS	×	CGPS	×	×	×
SR ₁	×	CGPS	CGPS	×	×	CGPS	×	×
SR ₂	×	CGPS	CGPS	×	×	CGPS	×	×
SR ₃	×	CGPS	CGPS	×	×	CGPS	×	×
SR ₄	×	CGPS	CGPS	×	×	CGPS	×	×
AR ₁ ^F	×	CGPS	CGPS	×	CGPS	×	×	×
AR ₂ ^F	×	CGPS	CGPS	×	CGPS	×	×	×
AR ₃ ^F	×	CGPS	CGPS	×	CGPS	×	×	×
AR ₄ ^F	×	CGPS	CGPS	×	CGPS	×	×	×
AR ₁ ^A	×	CGPS	CGPS	×	×	×	×	×
AR ₂ ^A	×	CGPS	CGPS	×	×	×	×	×
AR ₃ ^A	×	CGPS	CGPS	×	×	×	×	×
AR ₄ ^A	×	CGPS	CGPS	×	×	×	×	×
OR	CGPS	CGPS	CGPS	CGPS	CGPS	CGPS	×	CGPS
NBR	CGPS	CGPS	CGPS	CGPS	CGPS	×	×	×

Table 6

Comparison of the reductions in terms of their satisfaction of new agent principles (P10–P17)

7. New Agent Principles

In this section, we introduce eight new principles to distinguish between agent semantics. Principle 10 says that if more agents adopt an argument that is accepted, this does not affect the extension.

Principle 10 (AgentAdditionPersistence). An agent semantics δ satisfies AgentAdditionPersistence iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, every $E \in \delta(AAF)$, every $\alpha \in \mathcal{S}$, and every $a \in E$, we have $E \in \delta(\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \cup (a, \alpha) \rangle)$.

Principle 11 reflects the same idea as Principle 10, but is based on the assumption that a is accepted in all extensions.

Principle 11 (AgentAdditionUniversalPersistence). An agent semantics δ satisfies AgentAdditionUniversalPersistence iff, for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, every $E \in \delta(AAF)$, every $\alpha \in \mathcal{S}$, and every $a \in E$, we have $\forall E \in \delta(\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \cup (a, \alpha) \rangle), a \in E$.

Proposition 13. OR satisfies Principle 10 and Principle 11 for all the semantics.

Proof. Assume an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, $OR(AAF) = \langle \mathcal{A}', \rightarrow' \rangle$. For any extension $E \in \delta(AAF)$, $\forall a \in E$, there exists an agent α such that $a \sqsubseteq \alpha$. By definition, we find that any argument in the extension has at least one agent, so attaching more agents to AAF will not affect $OR(AAF)$. Thus, OR satisfies Principle 10 and Principle 11 for all the semantics. $\square$

Principle 12 reflects a principle we expect to hold for all agent semantics. It expresses anonymity: permuting the agents does not affect the extensions. This is analogous to language independence for arguments as defined by Baroni and Giacomin [10].

Principle 12 (PermutationPersistence). An agent semantics δ satisfies PermutationPersistence iff, for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and $AAF' = \langle \mathcal{A}, \rightarrow, \mathcal{S}', \sqsubset' \rangle$, where $\mathcal{S}$ and $\mathcal{S}'$ are two different ordered sets with common elements, we have $\delta(AAF) = \delta(AAF')$.

Principle 13 reflects that if the arguments of two agents do not attack one another, we can merge these agents into one single agent. This does not hold for agent defense semantics because new agent defenses may be created.

Principle 13 (MergeAgent). An agent semantics δ satisfies MergeAgent iff, for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, and for any $\alpha, \beta \in \mathcal{S}$ such that, for $\forall a \in \mathcal{A}_\alpha$ and $\forall b \in \mathcal{A}_\beta$, a does not attack b and b does not attack a , we have AAF' obtained by changing $\forall a \sqsubset \alpha$ to $a \sqsubset \beta$ for all $\delta(AAF) = \delta(AAF')$.

Principle 14 reflects that if two agents have the same arguments, we can remove one of these agents without changing the extensions. This represents the opposite of social semantics, where the number of agents makes a difference.

Principle 14 (RemovalAgentPersistence). An agent semantics δ satisfies RemovalAgentPersistence iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and for any $\alpha, \beta \in \mathcal{S}$ such that $\mathcal{S}_\alpha = \mathcal{S}_\beta$, we have $\delta(\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle) = \delta(\langle \mathcal{A}, \rightarrow, \mathcal{S} \setminus \alpha, \sqsubset \setminus \sqsubset_\alpha \rangle) = \delta(\langle \mathcal{A}, \rightarrow, \mathcal{S} \setminus \beta, \sqsubset \setminus \sqsubset_\beta \rangle)$.

Principle 15 is inspired by social agent semantics. It states that for two argumentation frameworks with the same arguments and attacks, if for every argument the number of agents holding that argument is the same, then the extensions are the same.

Principle 15 (AgentNumberEquivalence). An agent semantics δ satisfies AgentNumberEquivalence iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and $AAF' = \langle \mathcal{A}, \rightarrow, \mathcal{S}', \sqsubset' \rangle$, for $\forall a \in \mathcal{A}$, $|\mathcal{S}_a| = |\mathcal{S}'_a|$, we have $\delta(AAF) = \delta(AAF')$.

Principle 16 is inspired by agent reduction semantics. It states that if the set of the arguments of an agent is conflict-free, then there is an extension containing those arguments.

Principle 16 (ConflictFreeInvolvement). An agent semantics δ satisfies ConflictFreeInvolvement iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, and for $\forall \alpha \in \mathcal{S}$, it is the case that $\mathcal{A}_\alpha$ is conflict-free, there is an E , and we have $\mathcal{A}_\alpha \subseteq E$.

Principle 17 is inspired by OrphanReduction semantics. It states that if we have arguments that are not held by any agent, then they can be removed from the framework without affecting the extensions.

Principle 17 (RemovalArgumentPersistence). An agent semantics δ satisfies RemovalArgumentPersistence iff for every $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, and for $\nexists \alpha \in \mathcal{S}$ and $a \sqsubset \alpha$, we have $\delta(\langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle) = \delta(\langle \mathcal{A} \setminus a, \rightarrow \setminus \rightarrow_a, \mathcal{S}, \sqsubset \rangle)$.

Most of the principles are independent — in particular, Principle 3 (P3 for short), P5, P12, P13, P14, P15, P16, and P17. However, some principles have inner relationships among themselves. For example, if a semantic satisfies P2, then it must satisfy P1; we denote this observation as $P2 \Rightarrow P1$. The other observations are: $P4 \Rightarrow P8 \Rightarrow P9$, $P6 \Rightarrow P7$, and $P10 \Rightarrow P11$.

As a result, in Table 6, all agent semantics satisfy P12. Perhaps surprisingly, both social agent semantics and agent reduction semantics do not satisfy P10, while trivial reduction semantics, social agent semantics and agent filtering semantics satisfy P13. Moreover, all agent semantics except social agent

semantics satisfy P14. No semantics satisfies P16. As expected, only OrphanRemoval satisfies P17. The only semantics that are not distinguished concern the use of different preference reductions or different Dung semantics. To distinguish between them, the principles proposed in preference-based argumentation and in Dung's semantics can be used. In that sense, the principle-based analysis in this article complements the principle-based analysis in other areas.

8. Related Work

Our research is related to different branches of the literature on formal argumentation.

There are other semantics variants that adapt the notion of defense for abstract argumentation frameworks, such as weak defense for weak admissibility semantics [12], but they are not agent-based. Many extensions of abstract argumentation frameworks have been proposed, together with more general notions of acceptance. Some example additions to the basic argumentation frameworks are preferences [29], support relations [19, 46], abstract dialectical frameworks [15], and higher-order relations [18]. Abstract dialectical frameworks are discussed in detail in the first volume of the Handbook of Formal Argumentation [9] while other extensions are discussed in the second volume of the Handbook [24]. However, within these settings, we are not aware of other approaches that adapt Dung's basic concepts directly with the additional elements, as we have done with individual and collective agent defense.

There is a striking similarity at the abstract level between preference-based argumentation and support in bipolar argumentation — both can be seen as social reductions. One of the most closely related work comes from the field of social agent semantics. Leite and Martins [32] introduced an abstract model of argumentation in which agents can vote for or against an argument. They defined an abstract argumentation framework as a triple $\langle \mathcal{A}, \mathcal{R}, \mathcal{V} \rangle$, where $\mathcal{V} \rightarrow N \times N$ is a total function that gives, for each argument, the number of positive (Pro) and negative (Con) votes. Our article, on the other hand, considers only positive votes. Caminada and Pigozzi [17] captured the notion that individual members need to defend the collective decision to reach a compatible outcome. They proposed to address judgment aggregation by combining different individual evaluations of the situation, represented by an argumentation framework.

From the agent perspective, there is a choice between 1) combining the individual agents' frameworks into a common framework by voting on the existence of arguments and attacks, or 2) making it so that agents can agree on the framework and vote on the extensions. In this article, we have considered and compared both approaches. Additionally, Rienstra et al. [40] consider the case where the agents may have different semantics. For example, one agent uses grounded semantics while another uses preferred semantics. Furthermore, Kontarinis and Toni [31] analyse the identification of malicious behavior by agents using bipolar argumentation frameworks which, together with the work of Panisson et al. [36], may inspire work on agent reduction semantics based on trustfulness.

When considering agents' knowledge, epistemology, beliefs, and trust assessment, filtering semantics come into play. Hunter et al. [28] take an epistemic approach to probabilistic argumentation, where arguments are believed or not believed to different degrees, thus providing an alternative to the subtle standard Dung framework. Fazzinga et al. [23] have proposed a trust-aware abstract argumentation framework (T-AAF) and an agent-aware abstract argumentation framework (Ag-AAF). They extend traditional frameworks by associating arguments with the agents who hold them and, in T-AAFs, assigning trust scores to these agents. This setup allows for trust-based filtering, enabling analysis of argument robustness by excluding arguments from less trusted agents. Ag-AAFs, in contrast, evaluate argument

sets based solely on agent identity, thereby offering insights into argument robustness without relying on trust metrics. This approach could complement our work by providing a method for dynamically assessing argument acceptance based on agent reliability, which allows for flexible analysis in trust-sensitive environments. Yu et al. [51] have developed a context-based argumentation system (CAS) that supports consensus-building by enabling agents to prioritize norms and values in specific contexts, dynamically adapting to shifts in argument priorities. CAS's flexibility in adjusting argument preferences based on evolving socio-cultural or legal contexts contrasts with static aggregation methods, making it suitable for multi-agent scenarios that require context-sensitive adaptations. Lastly, Sakama [43] introduces argumentation frameworks with beliefs (AFB), where agents can hold nested beliefs about arguments, thus allowing for complex belief states, such as inner conflict, within argumentation. This belief-driven approach adds a unique perspective on agent viewpoints, complementing formal acceptance-based frameworks by emphasising the internal belief dynamics of agents.

In their exploration of multi-agent argumentation and dialogue, Arisaka et al. [3] have extended Dung's abstract argumentation framework by assigning arguments to agents and by incorporating agent interaction and dialogue. Their approach centers on the concept of conditional acceptance, where an agent evaluates arguments not in isolation but relative to the trustworthiness of the sources (other agents) supporting these arguments. To formalize conditional and multi-agent argumentation, the authors employ the theory of input/output argumentation [8], also referred to as multi-sorted argumentation [41]. They distinguish between individual and collective acceptance — individual acceptance is determined by each agent's trust-based evaluation of arguments, while collective acceptance involves aggregating revealed arguments to determine global acceptability. Additionally, agents can strategically choose to hide certain arguments from others, revealing information selectively to shape dialogue outcomes. An external observer (e.g., a judge) may accept arguments that individual agents do not, applying an independent evaluation that incorporates or modifies agents' disclosures. In this regard, Arisaka et al.'s [3] concept of collective acceptance parallels our agent reduction semantics, where individual agent perspectives are aggregated to define collective extensions at a global level.

Similar approaches have been applied to abstract argumentation semantics [10, 45], preference-based argumentation [29], and bipolar argumentation [46, 48]. The principles introduced here differ substantially from those in the existing literature as they focus specifically on the agent aspect and are orthogonal to the principles that characterize Dung's semantics. In addition, we have added some principles that reflect important properties of agent semantics and that can be used to guide the development of future agent semantics. More specifically, by combining ideas from earlier work in abstract argumentation, our principle-based analysis has revealed several original insights. For example, a new twist was given to the fundamental role of defense and reinstatement in Dung's theory in the context of agent defense semantics.

9. Future Work

The attack-defense (AD) framework [34] has been recently proposed as a novel approach to abstract argumentation. It aims to address limitations in traditional Dung-style frameworks by redefining defense relations through a structured attack-defense (x, y, z) triple. This new framework captures complex interactions between arguments, enabling context-sensitive argumentation that cannot be represented in standard abstract frameworks. As future work, this AD framework opens the path to a broader unification of

argumentation theories by incorporating non-Dung notions of defense. Various extended frameworks—weak admissibility, bipolar argumentation, agent-based argumentation, value or preference-based frameworks, among others—can all be represented within AD semantics, illustrating a recurring pattern of defense redefinitions. This flexibility allows the AD framework to serve as a generalizable tool for reductions from bipolar argumentation frameworks, abstract agent argumentation, preference argumentation frameworks, and value argumentation frameworks, achieving integrations that are not feasible within traditional AF. Consequently, the AD framework not only enhances computational and expressive capacities but also lays the foundation for a more unified and extended understanding of defense, paving the way for significant future developments in formal argumentation.

One of the main challenges in formal argumentation is bridging the gap between abstract argumentation and dialogue [37]. To help address this gap, Caminada [16] introduced semantics for abstract argumentation that are suitable for structured discussions. However, implementing abstract argumentation within dynamic agent dialogues remains an open problem. The development of a comprehensive framework for argumentation as dialogue is a crucial direction for future research. Compared to the well-established formal study of argumentation-based inference, argumentation-based dialogue is still less developed, consisting primarily of diverse approaches and isolated systems with few unifying theories or general frameworks [37].

In our work, we explored abstract agent argumentation, introducing multiple semantics. However, the development of a structured theory of agent argumentation remains an open problem. Although extended abstract argumentation frameworks increase expressive capacity, further study is needed on the direct construction of these extensions from a knowledge base—including elements like agents, support relations, and numerical values or weights. Integrating agents more directly at the structured level seems particularly natural. For instance, systems like Jiminy [33] employ multiple stakeholders, each with distinct knowledge bases, to handle dilemmas and conflicts. Here, the argumentation engine can either combine individual arguments from each stakeholder to form a collective framework or merge the knowledge bases first before constructing an argumentation framework. Both approaches raise new questions about how to achieve coherent integration in structured argumentation settings.

As always with the principle-based approach, we can introduce more semantics. For example, we can combine the ideas of the four classes, guided by the existing principles. We can also study more principles. Or we can find relations with other branches of logic and reasoning such as axiomatic approaches in social choice. Moreover, we can try to use the principles discussed to address the standard challenges of abstract argumentation such as relating the abstract model to more structured forms of argumentation, and applying abstract argumentation to legal reasoning.

10. Summary

Our aim in this work has not been to prescribe a single “best” interpretation of agent argumentation semantics, but to provide a general, principle-based analysis of options that are natural to consider. As is common in the principle-based approach to argumentation semantics, we have selected principles that distinguish between different agent semantics, particularly the new principles that capture properties specific to the role of agents. These principles help us understand existing approaches, identify trade-offs, and guide the development of future semantics.

Combining ideas from earlier work in abstract argumentation, our analysis reveals several original insights. For instance, we give a new perspective on the fundamental role of defense and reinstatement

in Dung’s theory when extended to agent defense semantics. We also introduce a new variant of SCC-recursiveness for agent defense semantics. Since other approaches in our study are based on reductions, the traditional SCC-recursive algorithms for abstract argumentation can be reused in those cases.

An important feature of our framework is that it accommodates a variety of agent semantics. The priority or hierarchy among semantics, and the choice of which one to use, depend on the intended application, and they can also be combined. For example, one can use both filtering and agent defense: filtering to remove unknown arguments or attacks, and agent defense to support an argument put forward by a particular agent. The principles serve as a decision aid in this process. For instance, Principle 17 (Removal Argument Persistence) may lead to the use of agent filtering semantics for a given task. In this way, our framework not only organizes existing work but also provides guidance on selecting principled, application-driven semantics and lays the foundation for further research in this area.

For example, a new twist is given to the fundamental role of defense and reinstatement in Dung’s theory, within the context of agent defense semantics.

Finally, the future work section shows that our formal framework not only serves as a tool for organizing existing work in the area, but also provides a solid foundation for further work in this direction.

Acknowledgments

We thank the Luxembourg National Research Fund (FNR) for its support through the project Deontic Logic for Epistemic Rights (DELIGHT) (OPEN O20/14776480). Leendert van der Torre acknowledges the financial support of the FNR through the following projects: The Epistemology of AI Systems (EAI) (C22/SC/17111440), DJ4ME – A DJ for Machine Ethics: the Dialogue Jiminy (O24/18989918/DJ4ME), Logical Methods for Deontic Explanations (LoDEx) (INTER/DFG/23/17415164/LoDEx) and Symbolic and Explainable Regulatory AI for Finance Innovation (SERAFIN) (C24/19003061/SERAFIN). Liuwen Yu thanks the FNR for their support through the SERAFIN project (C24/19003061/SERAFIN) and the University of Luxembourg for the Marie Speyer Excellence Grant for the project Formal Analysis of Discretionary Reasoning (MSE-DISCREASON).

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11. Appendix

11.1. TR

Proposition 14. TR satisfies Principles 10, 11, 12, 13, 14, and 15 for all the semantics.

Proof. Semantics under TR does not concern the concept of agent, thus, they all satisfy Principles 10, 11, 12, 13, 14, and 15. $\square$

Proposition 15. TR does not satisfy Principles 16 and 17.

Proof. We use a counterexample to prove that TR does not satisfy Principles 16 and 17, as shown in Figure 8. The complete, grounded, and stable semantics under TR is $\{a\}$. $\square$

11.2. Sem_1 and Sem_2

Proposition 16 ($Sem_1, Sem_2 \times P_1, P_2$). All four kinds of Dung semantics under admissibility₁ and/or admissibility₂ satisfy Principle 1 and Principle 2.

Proof. Straightforward by definition. $\square$

Proposition 17 ($Sem_1 \times P_6$). The grounded, complete, and preferred semantics under admissibility₁ satisfy Principle 6.

Proof. Straightforward by definition. $\square$

Proposition 18 ($Sem_2 \times P_7$). The grounded, complete, and preferred semantics under admissibility₂ satisfy Principle 7.

Proof. Straightforward by definition. $\square$

Proposition 19 ($Sem_1 \times P_7$). The grounded, complete, and preferred semantics under admissibility₁ satisfy Principle 7.

Proof. It suffices to show that, given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \square \rangle$, for every $E \subseteq \mathcal{A}$ and $c \in \mathcal{A}$, if $E \text{ defend}_1 c$ then $E \text{ defend}_2 c$. This is straightforward by definition. $\square$

Proposition 20 ($Sem_2 \times P_6$). None of the four kinds of Dung semantics under admissibility₂ satisfy Principle 6.

Proof. Consider the AAF in Figure 17. It is easy to see that, for every $E \subseteq \mathcal{A}$, if E is admissible₁

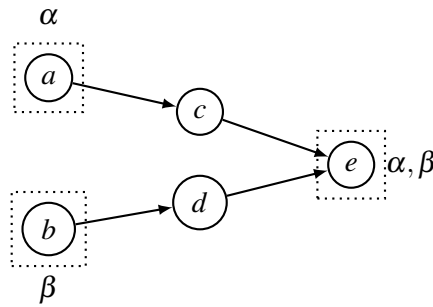


Fig. 17. Agent defense

then $e \notin E$ (since otherwise E cannot defend₁ itself). On the other hand, it is easy to see that $\{a, b, e\}$ is the grounded extension of the above AAF under admissibility₂, thus the grounded semantics under admissibility₂ does not satisfy P_6 . Furthermore, since the grounded extension is the least complete extension, the complete semantics and preferred semantics under admissibility₂ do not satisfy P_6 either. The stable semantics under admissibility₂ is considered in Proposition 10. $\square$

Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $\mathcal{U} \subseteq \mathcal{A}$. For every $E \subseteq \mathcal{U}$ and $c \in \mathcal{U}$, we write “ $E \text{ defend}_i^{\mathcal{A}} c$ ” if $E \text{ defend}_i c$ in AAF and we write “ $E \text{ defend}_i^{\mathcal{U}} c$ ” if $E \text{ defend}_i c$ in $AAF \downarrow_{\mathcal{U}}$, where $i \in \{1, 2\}$.

Lemma 11.1. *Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $\mathcal{U} \subseteq \mathcal{A}$ be an unattacked set. For every $i \in \{1, 2\}$, every set $E \subseteq \mathcal{A}$, and every argument $c \in \mathcal{U}$, we have that $E \text{ defend}_i^{\mathcal{A}} c$ iff $E \cap \mathcal{U} \text{ defend}_i^{\mathcal{U}} c$.*

Proof. We only consider the case of $i = 1$. From left to right: assume $E \text{ defend}_1^{\mathcal{A}} c$. Then there must be an $\alpha \in \mathcal{S}_c$ such that for all arguments $b \in c^-$, there exists an $a \in E \cap A_\alpha$ such that $a \in b^-$. Now let $b \in c^- \cap \mathcal{U}$ be arbitrary. It follows that there exists an $a \in E \cap A_\alpha$ such that $a \in b^-$. Since $\mathcal{U}$ is unattacked and $b \in \mathcal{U}$, we have $a \in \mathcal{U}$ as well, i.e., $a \in (E \cap \mathcal{U}) \cap (A_\alpha \cap \mathcal{U})$. Since b is arbitrary, $E \text{ defend}_1^{\mathcal{U}} c$.

From right to left: assume $E \cap \mathcal{U} \text{ defend}_1^{\mathcal{U}} c$. Then there must be an $\alpha \in \mathcal{S}_c$ such that for all arguments $b \in c^- \cap \mathcal{U}$, there exists an $a \in E \cap (A_\alpha \cap \mathcal{U})$ such that $a \in b^-$. Since $\mathcal{U}$ is unattacked and $c \in \mathcal{U}$, we have $c^- \cap \mathcal{U} = c^-$. It follows that for all arguments $b \in c^-$, there exists an $a \in E \cap A_\alpha$ such that $a \in b^-$. That is, $E \text{ defend}_1^{\mathcal{A}} c$. $\square$

Proposition 21 ($\text{Sem}_1, \text{Sem}_2 \times \text{P}_3$). Grounded semantics, complete semantics, and preferred semantics under admissibility₁ and/or admissibility₂ satisfy Principle 3, whereas stable semantics under admissibility₁ and/or admissibility₂ does not satisfy Principle 3.

Proof. Since stable semantics under admissibility₁ and/or admissibility₂ is the same as that in abstract argumentation frameworks, the second half follows from [10] directly.

For the first half, we first consider complete semantics. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $\mathcal{U} \subseteq \mathcal{A}$ be an unattacked set. If $E \subseteq \mathcal{A}$ is a complete extension of AAF , we show that $E \cap \mathcal{U}$ is a complete extension of $AAF \downarrow_{\mathcal{U}}$. It is easy to see that $E \cap \mathcal{U}$ is conflict-free. For every $c \in E \cap \mathcal{U}$, since $E \text{ defend}_i^{\mathcal{A}} c$, we have that $E \cap \mathcal{U} \text{ defend}_i^{\mathcal{U}} c$ by Lemma 11.1. For every $c \in \mathcal{U} \setminus E$, since E does not $\text{defend}_i^{\mathcal{A}} c$, we have that $E \cap \mathcal{U}$ does not $\text{defend}_i^{\mathcal{U}} c$ by Lemma 11.1.

On the other hand, if $E \subseteq \mathcal{U}$ is a complete extension of $AAF \downarrow_{\mathcal{U}}$, then by Lemma 11.1 it is easy to see that E is still admissible_i in AAF . For every $j \in \mathbb{N}$, we inductively define a set $E_j \subseteq \mathcal{A}$ as follows: (1) $E_0 = E$; (2) $E_{n+1} = E_n \cup \text{defend}_i^{\mathcal{A}}(E_n)$. Let $E^* = \bigcup_{j \in \mathbb{N}} E_j$. We show that $E^* \cup \mathcal{U} = E$ and E^* is a complete extension of AAF . For the former, using Lemma 11.1, we can show that $E_j \cap \mathcal{U} = E$ for every $j \in \mathbb{N}$. The case of $j = 0$ is trivial. Suppose $E_n \cap \mathcal{U} = E$, then $E_{n+1} \cap \mathcal{U} = (E_n \cap \mathcal{U}) \cup (\text{defend}_i^{\mathcal{A}}(E_n) \cap \mathcal{U})$. By Lemma 11.1, $\text{defend}_i^{\mathcal{A}}(E_n) \cap \mathcal{U} = \text{defend}_i^{\mathcal{U}}(E_n \cap \mathcal{U}) = \text{defend}_i^{\mathcal{U}}(E) \subseteq E$ (note that E is a complete extension of $AAF \downarrow_{\mathcal{U}}$). Thus $E_{n+1} \cap \mathcal{U} = E$. Therefore $E^* \cup \mathcal{U} = E$. For the latter, since E is admissible_i in AAF , it is easy to prove that E^* is a complete extension of AAF . This completes the proof for complete semantics.

Next we consider grounded semantics. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $\mathcal{U} \subseteq \mathcal{A}$ be an unattacked set. Suppose $E \subseteq \mathcal{A}$ is the grounded extension of AAF . Then $E \cap \mathcal{U}$ must be the grounded extension of $AAF \downarrow_{\mathcal{U}}$; otherwise, the grounded extension of $AAF \downarrow_{\mathcal{U}}$ would be $E' \subset E \cap \mathcal{U}$. Consider the set $(E')^*$ defined as in the last paragraph. Then $(E')^* \cup \mathcal{U} = E'$ and $(E')^*$ is a complete extension of AAF . It follows that $E \not\subseteq (E')^*$, contradicting that E is the grounded extension of AAF . On the other hand, suppose that $E \subseteq \mathcal{U}$ is the grounded extension of $AAF \downarrow_{\mathcal{U}}$. Consider the set E^* defined as in the last paragraph. It is a complete extension of AAF . Let $E' \subseteq \mathcal{A}$ be an arbitrary complete extension of AAF . We know that $E' \cap \mathcal{U}$ is a complete extension of $AAF \downarrow_{\mathcal{U}}$, thus $E \subseteq E' \cap \mathcal{U} \subseteq E'$. By induction on j , we can prove that $E_j \subseteq E'$ for every $j \in \mathbb{N}$. Thus $E^* \subseteq E'$. Since E' is arbitrary, E^* is the grounded extension of AAF .

Finally we consider preferred semantics. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $\mathcal{U} \subseteq \mathcal{A}$ be an unattacked set. Suppose that $E \subseteq \mathcal{A}$ is a preferred extension of AAF . Since preferred extensions are maximal complete extensions, E is also a complete extension of AAF . Thus $E \cap \mathcal{U}$ is a complete extension of $AAF \downarrow_{\mathcal{U}}$. We show that $E \cap \mathcal{U}$ is a preferred extension of $AAF \downarrow_{\mathcal{U}}$. Suppose not. Then there is an $F \subseteq \mathcal{U}$ such that $E \cap \mathcal{U} \subset F$ and F is a complete extension of $AAF \downarrow_{\mathcal{U}}$. Consider the set $F \cup E$. We show that $F \cup E$ is admissible, which contradicts that E is maximal admissible. We first show that $F \cup E$ is conflict-free. Suppose not. Since E and F are conflict-free and $\mathcal{U}$ is unattacked, the only possibility is that there exist $a \in F \setminus E$ and $b \in E \setminus F$ such that a attacks b . Since E defend _{ℓ} ^{$\mathcal{A}$} b , there must be a $c \in E$ such that c attacks a . Note that $a \in F \subseteq \mathcal{U}$ and $\mathcal{U}$ is unattacked, thus $c \in \mathcal{U} \cap E \subset F$, which implies that F is not conflict-free, a contradiction! It remains to be shown that $F \cup E$ can defend itself. This is trivial in view of Lemma 11.1.

On the other hand, suppose $E \subseteq \mathcal{U}$ is a preferred extension of $AAF \downarrow_{\mathcal{U}}$. Consider the set E^* defined as before. E^* is a complete extension of AAF and $E^* \cup \mathcal{U} = E$. Since $\mathcal{A}$ is finite, there must be a maximal complete extension (preferred extension) F of AAF such that $F \supseteq E^*$. Consider the set $F \cap \mathcal{U}$. We know that $F \cap \mathcal{U}$ is a preferred extension of $AAF \downarrow_{\mathcal{U}}$ by the previous paragraph. Note that $E \subseteq F \cap \mathcal{U}$, thus $E = F \cap \mathcal{U}$. $\square$

Proposition 22 (Sem₁, Sem₂ $\times$ P₅). Grounded semantics, complete semantics, and preferred semantics under admissibility₁ and/or admissibility₂ do not satisfy Principle 5, whereas stable semantics under admissibility₁ and/or admissibility₂ does satisfy Principle 5.

Proof. Since stable semantics under admissibility₁ and/or admissibility₂ is the same as that in abstract argumentation frameworks, the second half follows from [12] directly.

For the first half of the lemma, consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 11.

Under both admissibility₁ and admissibility₂, it is clear that $\{a\}$ is not only the grounded extension of the above AAF but also its complete and preferred extensions.

Let us consider the complete semantics: the set $\{a\}$ is a complete extension of the AAF and the set $\{c\}$ is a complete extension of $AAF^{\{a\}} = AAF \downarrow_{\{c\}}$. However, $\{a, c\}$ is not a complete extension of the AAF . $\square$

Lemma 11.2. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, a set $E \subseteq \mathcal{A}$ and an SCC $S \in SCCS_{AAF}$, we have the following for all $a \in S$:

- (1) For each $i \in \{1, 2\}$, we have that $a \in AUP_{AAF}^i(S, E)$ iff $\forall b \in E \setminus S$ it holds that $b \not\rightarrow a$.
- (2) We have that $a \in AU_{AAF}^1(S, E)$ iff $\forall b \in E \setminus S$ it holds that $b \not\rightarrow a$ and there exists $\alpha \in \mathcal{S}_a$ such that $\forall b \in \mathcal{A} \setminus S$ with $b \rightarrow a$ there exists $c \in E \cap \mathcal{A}_\alpha$ such that $c \rightarrow b$.
- (3) We have that $a \in AU_{AAF}^2(S, E)$ iff $\forall b \in E \setminus S$ it holds that $b \not\rightarrow a$ and $\forall b \in \mathcal{A} \setminus S$ with $b \rightarrow a$, there exists $\alpha \in \mathcal{S}_a$ and $c \in E \cap \mathcal{A}_\alpha$ such that $c \rightarrow b$.

Proposition 23 (Sem₁ $\times$ P₉). Complete, preferred and grounded semantics under admissibility₁ do not satisfy Principle 9.

Proof. Consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 17: Under admissibility₁, it is clear that $\{a, b\}$ is not only the grounded extension of the above AAF but also its complete and preferred extensions.

We first consider grounded semantics. Suppose, toward a contradiction, that there is a function $\mathcal{G}$ as described in Principle 9. Thus $\mathcal{G}(AAF, \mathcal{A}) = \{\{a, b\}\}$ (1). Now, let $E \in \mathcal{G}(AAF, \mathcal{A})$. There are five

SCCs in the AAF, i.e., $S_1 = \{a\}$, $S_2 = \{b\}$, $S_3 = \{c\}$, $S_4 = \{d\}$ and $S_5 = \{e\}$. By (1) we know that $E \cap S_1 = \{a\}$ and $E \cap S_2 = \{b\}$. Now consider the SCC S_3 . Since $\text{AUP}_{AAF}^2(S_3, E) = \text{AU}_{AAF}^2(S_3, E) = \emptyset$, it holds that $E \cap S_3 = \emptyset$. For the same reason, $E \cap S_4 = \emptyset$. However, S_5 remains to be considered. Note that $\text{AUP}_{AAF}^2(S_5, E) = \text{AU}_{AAF}^2(S_5, E) = \{c\}$. So $E \cap S_5 \in \mathcal{G}(AAF \downarrow_{\{c\}}, \{c\})$. Note that $\mathcal{G}(AAF \downarrow_{\{c\}}, \{c\})$ must be the grounded extension of $AAF \downarrow_{\{c\}}$ (under admissibility₁), which is $\{c\}$. So $E = \{a, b, c\}$, contradicting (1).

Complete and preferred semantics can be shown in a similar way. $\square$

Proposition 24. Stable semantics under admissibility₁ and/or admissibility₂ satisfies Principle 8 and Principle 9.

Proof. We define the function $\mathcal{G}$ as follows. For any $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and $C \subseteq \mathcal{A}$, it holds that $E \in \mathcal{G}(AAF, C)$ if and only if E is a stable extension in $\langle \mathcal{A}, \rightarrow \rangle$. By Proposition 32 in [11], it is straightforward to show that $\mathcal{G}$ satisfies the properties in Principle 8 and Principle 9. $\square$

Proposition 25 ($\text{Sem}_2 \times \text{P}_8$). Complete, grounded and preferred semantics for admissibility₂ do not satisfy Principle 8.

Proof. Consider $AAF_1 = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset_1 \rangle$ and $AAF_2 = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset_2 \rangle$ in Figure 11.2.

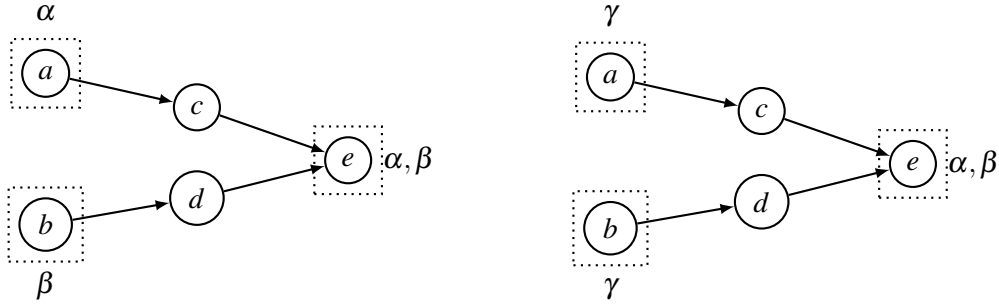


Fig. 18. Two AAFs: AAF_1 (on the left) and AAF_2 (on the right)

Let us consider, for example, the complete semantics under admissibility₂. We denote the five SCCs in AAF_1 (or AAF_2) as: $S_1 = \{a\}$, $S_2 = \{b\}$, $S_3 = \{c\}$, $S_4 = \{d\}$ and $S_5 = \{e\}$. Under admissibility₂, it is clear that $E = \{a, b, e\}$ is a complete extension of AAF_1 . Thus $\forall i \in \{1, \dots, 5\}$:

$$E \cap S_i \in \mathcal{G}(AAF_1 \downarrow_{\text{AUP}_{AAF_1}^1(S_i, E)}, \text{AU}_{AAF_1}^1(S_i, E)).$$

We can also verify that $\forall i \in \{1, \dots, 5\}$:

$$E \cap S_i \in \mathcal{G}(AAF_2 \downarrow_{\text{AUP}_{AAF_2}^1(S_i, E)}, \text{AU}_{AAF_2}^1(S_i, E)).$$

By Principle 8, it must be that $E \in \mathcal{G}(AAF_2, \mathcal{A})$. But E is not a complete extension of AAF_2 under admissibility₂, a contradiction! $\square$

Proposition 26. Sem_1 does not satisfy Principle 13 for all the semantics.

Proof. Figure 2 is a counterexample to prove Sem_1 does not satisfy Principle 13. $\square$

Proposition 27. Sem_1 does not satisfy Principle 15 for all the semantics.

Proof. We provide a counterexample, illustrated in Figure 19. $\square$

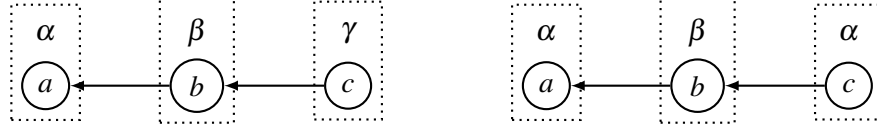


Fig. 19. Sem_1 does not satisfy Principle 15.

Proposition 28. Sem_1 does not satisfy Principle 16 for all the semantics.

Proof. A counterexample is Figure 2. $\square$

Proposition 29. Sem_1 does not satisfy Principle 16 for all the semantics.

Proof. A counterexample is Figure 8. $\square$

Proposition 30.

11.3. $\text{SR}_1 - \text{SR}_4$

Proposition 31 ($\text{SR}_1 \times \text{P}_1, \text{P}_2$). All four kinds of Dung semantics for SR_1 do not satisfy Principle 1. Thus, they do not satisfy Principle 2 either.

Proof. A counterexample is Figure 8. $\square$

Proposition 32 ($\text{SR}_2 \times \text{P}_1$). All four kinds of Dung semantics for SR_2 satisfy Principle 1.

Proof. Let an $\text{AAF} = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \square \rangle$ be given. It suffices to show that for any $a, b \in \mathcal{A}$, if there is no attack between them in $\text{SR}_2(\text{AAF})$, then there is no attack between them in AAF . Suppose not. Without loss of generality, we assume that a attacks b in AAF . If $|\mathcal{S}(b)| \not\geq |\mathcal{S}(a)|$, then $b \not\succeq a$. Thus it must be that a attacks b in $\text{SR}_2(\text{AAF})$, a contradiction! If $|\mathcal{S}(b)| > |\mathcal{S}(a)|$, we consider two cases. If b attacks a in AAF , then it must be that b attacks a in $\text{SR}_2(\text{AAF})$, a contradiction. If b does not attack a in AAF , then since $b \succ a$, it must be that b attacks a in $\text{SR}_2(\text{AAF})$, a contradiction! $\square$

Proposition 33 ($\text{SR}_3 \times \text{P}_1$). All four kinds of Dung semantics for SR_3 satisfy Principle 1.

Proof. Let an $\text{AAF} = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \square \rangle$ be given. It suffices to show that for any $a, b \in \mathcal{A}$, if there is no attack between them in $\text{SR}_3(\text{AAF})$, then there is no attack between them in AAF . Suppose not. Without loss of generality, we assume that a attacks b in AAF . If b does not attack a in AAF , then a attacks b in $\text{SR}_3(\text{AAF})$, a contradiction! If b attacks a in AAF , since it must be either $|\mathcal{S}(b)| \not\geq |\mathcal{S}(a)|$ or $|\mathcal{S}(a)| \not\geq |\mathcal{S}(b)|$, there must be an attack between a and b in $\text{SR}_3(\text{AAF})$, a contradiction! $\square$

Proposition 34 ($SR_4 \times P_1$). All four kinds of Dung semantics for SR_4 satisfy Principle 1.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given. It suffices to show that for any $a, b \in \mathcal{A}$, if there is no attack between them in $SR_4(AAF)$, then there is no attack between them in AAF . This is easy, because if there is no attack between them in $SR_4(AAF)$, then there is no attack between them in $SR_2(AAF)$. $\square$

Proposition 35 ($SR_2 \times P_2$). All four kinds of Dung semantics for SR_2 do not satisfy Principle 2.

Proof. A counterexample is Figure 8. $\square$

Proposition 36 ($SR_3 \times P_2$). All four kinds of Dung semantics for SR_3 satisfy Principle 2.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given. It suffices to show that any admissible set $E \subseteq \mathcal{A}$ in $SR_3(AAF)$ is admissible in $\langle \mathcal{A}, \rightarrow \rangle$. Suppose, toward a contradiction, that E is not admissible in $\langle \mathcal{A}, \rightarrow \rangle$. By Proposition 33, it must be that there is a $c \in E$ such that E cannot defend c in $\langle \mathcal{A}, \rightarrow \rangle$. Thus, since E is conflict-free in $\langle \mathcal{A}, \rightarrow \rangle$, there must be a $b \in \mathcal{A} \setminus E$ such that b attacks c and a does not attack b for all $a \in E$ (in $\langle \mathcal{A}, \rightarrow \rangle$). In particular, c does not attack b (in $\langle \mathcal{A}, \rightarrow \rangle$). Therefore, by definition, b also attacks c in $SR_3(AAF)$. But for all $a \in E$, it holds that a does not attack b in $SR_3(AAF)$. So E cannot defend c in $SR_3(AAF)$, a contradiction! $\square$

Proposition 37 ($SR_4 \times P_2$). Complete semantics, preferred semantics and stable semantics for SR_4 do not satisfy Principle 2, whereas the grounded semantics for SR_4 does satisfy Principle 2.

Proof. For the first half, we have the following counterexample: Note that $\{b\}$ a complete, preferred, or



Fig. 20. SR_4 does not satisfy Principle 2.

stable extension of $SR_4(AAF)$, but it is not admissible in AAF .

For the second half of the lemma, let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let $E \subseteq \mathcal{A}$ be the grounded extension of $SR_4(AAF)$. Since E is conflict-free by Proposition 34, we only need to show that E can defend all its members in AAF . Let D be the characteristic function of $SR_4(AAF)$, thus $E = \bigcup_{i=1, \dots, \infty} D^i(\emptyset)$. We prove, by induction on the value of i , that E can defend $D^i(\emptyset)$ in AAF for $i = 1, \dots, \infty$. If $i = 1$, let $c \in D^1(\emptyset) = D(\emptyset)$ be arbitrary. For any argument b such that b attacks c and c does not attack a in AAF , it must be that b attacks c in $SR_4(AAF)$ by definition. Thus, such b does not exist, so E can defend $D^1(\emptyset)$ in AAF . Now suppose E can defend $D^n(\emptyset)$ in AAF . Let $c \in D^{n+1}(\emptyset)$ be arbitrary. For any argument b such that b attacks c and c does not attack a in AAF , it must be that b attacks c in $SR_4(AAF)$ by definition. Since $c \in D^{n+1}(\emptyset) = D(D^n(\emptyset))$, there must be an $a \in D^n(\emptyset)$ such that a attacks b in $SR_4(AAF)$. If a does not attack b in AAF , it must hold that b attacks a in AAF by definition. By IH, there is an $a' \in E$ such that a' attacks b in AAF , as desired. $\square$

Proposition 38 ($SR_2 \times P_3$). All four kinds of Dung semantics for SR_2 do not satisfy Principle 3.

Proof. A counterexample is Figure 21. $\square$

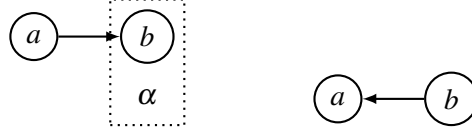


Fig. 21. An AAF (on the left) and $SR_2(AAF)$ (on the right).

Proposition 39 ($SR_4 \times P_3$). All four kinds of Dung semantics for SR_4 do not satisfy Principle 3.

Proof. A counterexample AAF is Figure 20. It is easy to see that $\{a\}$ is unattacked and is the grounded extension, complete extension, preferred extension and stable extension of $SR_4(AAF \downarrow_{\{a\}})$. But the grounded extension of $SR_4(AAF)$ is $\emptyset$ and $\{b\}$ is the complete, preferred and stable extension of $SR_4(AAF)$. $\square$

Proposition 40 ($SR_1, SR_2, SR_4 \times P_4$). All four kinds of Dung semantics for SR_1, SR_2 , and SR_4 do not satisfy Principle 4.

Proof. We first consider the case of SR_1 . Let $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, as shown in Figure 22. It is easy to see

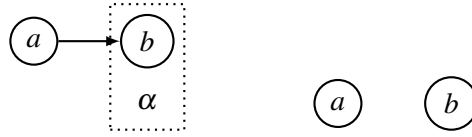


Fig. 22. An AAF (on the left) and $SR_1(AAF)$ (on the right).

that $\{a, b\}$ is not only the grounded extension of $SR_1(AAF)$, but also its complete, preferred and stable extension. Consider, for example, the grounded semantics for SR_1 . Suppose, toward a contradiction, that there is a function $\mathcal{G}$ as described in Principle 4. Then $\{a, b\} \in \mathcal{G}(AAF, \mathcal{A})$ (1). Now, let $E \in \mathcal{G}(AAF, \mathcal{A})$. There are two SCCs in AAF, i.e., $S_1 = \{a\}$ and $S_2 = \{b\}$. Since S_1 is the initial SCC, it follows that $UP_{AAF}(S_1, E) = U_{AAF}(S_1, E) = \{a\}$ for any E . Thus $E \cap S_1 \in \mathcal{G}(AAF \downarrow_{\{a\}}, \{a\})$. Since the grounded, complete, preferred and stable extensions of $SR_1(AAF \downarrow_{\{a\}})$ is $\{a\}$, we have $E \cap S_1 = \{a\}$. For S_2 , we have that $UP_{AAF}(S_2, E) = U_{AAF}(S_2, E) = \emptyset$, thus $E \cap S_2 = \emptyset$, contradicting (1). Complete, preferred, and stable semantics for SR_1 can be shown similarly.

We then consider the case for SR_2 . It is easy to see that $\{b\}$ is not only the grounded extension of $SR_2(AAF)$, but also a complete, preferred, and stable extension. But, by the same reasoning as above, $E \cap S_1 = \{a\}$ for any function $\mathcal{G}$ and $E \in \mathcal{G}(AAF, \mathcal{A})$.

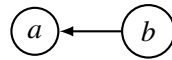
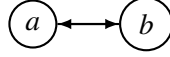


Fig. 23. $SR_2(AAF)$

The case for SR_3 remains to be considered. For the grounded semantics, note that $\emptyset$ is the grounded extension of $SR_4(AAF)$. But, by the same reasoning as above, $E \cap S_1 = \{a\}$ for any function $\mathcal{G}$ and

$E \in \mathcal{G}(AAF, \mathcal{A})$. For the complete, preferred, and stable semantics, note that $\{b\}$ is a complete, preferred, and stable extension of $SR_4(AAF)$. But, by the same reasoning as above, $E \cap S_2 = \emptyset$ for any function $\mathcal{G}$ and $E \in \mathcal{G}(AAF, \mathcal{A})$.

Fig. 24. $SR_4(AAF)$

Proposition 41 ($SR_2 \times P_3$). None of the four kinds of Dung semantics for SR_2 satisfy Principle 5.

Proof. Consider the counterexample in Figure 21. It is easy to see that $\{b\}$ is the grounded, complete, preferred and stable extension of $SR_2(AAF)$. Let $E = \{b\}$. Because AAF^E consists of the single point a , it follows that a is the grounded, complete, preferred and stable extension of $SR_2(AAF^E)$. However, $\{a, b\}$ is not consistent in $SR_2(AAF)$. $\square$

Proposition 42 ($SR_3 \times P_3$). None of the four kinds of Dung semantics for SR_3 satisfy Principle 5.

Proof. We first consider *the complete semantics*: let $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given, let E be a complete extension of $SR_3(AAF)$, and let E' be a complete extension of $SR_3(AAF^E)$. We show that $E \cup E'$ is a complete extension of $SR_3(AAF)$.

We first show that $E \cup E'$ is conflict-free in $SR_3(AAF)$. Suppose not. Then the only possibility is that there are $a \in E$ and $b \in E'$ such that a attacks b or b attacks a in $SR_3(AAF)$. The former case is impossible, because it follows that a attacks b in AAF and, thus, $b \in E^+$. In the latter case, since E defends a in $SR_3(AAF)$, there must be a $c \in E$ such that c attacks b in $SR_3(AAF)$. It follows that c attacks b in AAF . Thus $b \in E^+$, a contradiction!

We then show that $E \cup E'$ can defend itself in $SR_3(AAF)$. It is obvious that every $c \in E$ is defended by $E \cup E'$ in $SR_3(AAF)$. Let $c \in E'$ be arbitrary, and let b attack c in $SR_3(AAF)$. Since $E \cup E'$ is consistent in AAF , it follows that $b \in AAF^E$ or $b \in E^+$. In the former case, there is an $a \in E'$ such that a attacks b in AAF (since E' is a complete extension of $SR_3(AAF^E)$). In the latter case, there is an $a \in E$ such that a attacks b in AAF . Thus, it must be the case that either a attacks b or b attacks a in $SR_3(AAF)$. If b attacks a in $SR_3(AAF)$, then there must be an $a' \in E$ such that a' attacks b in $SR_3(AAF)$ because E defends a in $SR_3(AAF)$.

Finally, we show that $E \cup E'$ does not defend c for any $c \in \mathcal{A} \setminus (E \cup E')$. Suppose not. It follows that $c \notin E^+$; otherwise, there would be a $b \in E$ such that b attacks c in AAF . Therefore it must be the case that either b attacks c or c attacks b in $SR_3(AAF)$. In either case, there exists a $b' \in E$ such that b' attacks c in $SR_3(AAF)$. Since we assume that $E \cup E'$ defends c , there is an $a \in E \cup E'$ such that a attacks b' in $SR_3(AAF)$, contradicting the conflict-freeness of $E \cup E'$. For any argument $b \in E^*$ such that b attacks c in $SR_3(AAF^E)$ — thus, in $SR_3(AAF)$ — there is no $a \in E$ such that a attacks b in $SR_3(AAF)$ because that would imply that $b \in E^+$. Since we assume that $E \cup E'$ defends c in $SR_3(AAF)$, there must be an $a \in E'$ such that a attacks b in $SR_3(AAF)$ — thus, in $SR_3(AAF^E)$. It follows that E' defends c in $SR_3(AAF^E)$. Since E' is a complete extension of $SR_3(AAF^E)$, we obtain $c \in E'$, a contradiction!

We then consider *the grounded semantics*. Let $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given, let E be the grounded extension of $SR_3(AAF)$, and let E' be the grounded extension of $SR_3(AAF^E)$. It suffices to show that $E' = \emptyset$. We show the stronger claim that there is no argument c in $SR_3(AAF^E)$ that receives no attack (in

$SR_3(AAF^E)$). Suppose not. Then, for any $b \in \mathcal{A}$ such that b attacks c in $SR_3(AAF)$, we have $b \in E \cup E^+$. If $b \in E$, it follows that $c \in E^+$, contradicting that $c \in SR_3(AAF^E)$. If $b \in E^+$, then there is $a \in E$ such that a attacks b in $SR_3(AAF)$. So, E defends c in $SR_3(AAF)$. Since E is the grounded extension of $SR_3(AAF)$, we have $c \in E$, a contradiction!

For the preferred semantics, let E be a preferred extension of $SR_3(AAF)$ and let E' be a preferred extension of $SR_3(AAF^E)$. We show that $E' = \emptyset$. Otherwise, by the previous result, $E \cup E'$ is a complete extension of $SR_3(AAF)$ with $E \cup E' \supset E$.

Finally, for the stable semantics, suppose E is a stable extension of $SR_3(AAF)$ and E' is a stable extension of $SR_3(AAF^E)$. It is easy to see that $E' = \emptyset$ because $E^+ = \mathcal{A} \setminus E$. $\square$

Proposition 43 ($SR_4 \times P_5$). The complete, preferred and stable semantics for SR_4 do not satisfy Principle 5, whereas the grounded semantics for SR_4 does satisfy Principle 5.

Proof. For the first half, consider the counterexample in Figure 20.

It is easy to see that $\{b\}$ is a complete, preferred and stable extension of $SR_4(AAF)$. Let $E = \{b\}$, $SR_4(AAF^E)$ consists of the single point a . Thus $\{a\}$ is the complete, preferred and stable extension of $SR_4(AAF^E)$. However $\{b, a\}$ is not conflict-free in $SR_4(AAF)$.

For the second half, let $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given. let E be the grounded extension of $SR_4(AAF)$, and let E' be the grounded extension of $SR_4(AAF^E)$. It suffices to show that $E' = \emptyset$. We show the stronger claim that there is no argument c in $SR_4(AAF^E)$ that receives no attack (in $SR_4(AAF^E)$). Suppose not. Then, for any $b \in \mathcal{A}$ such that b attacks c in $SR_4(AAF)$, it holds that $b \in E \cup E^+$. We show that $b \notin E$ by proving the following claim:

For any $b \in E$, if b attacks c in $SR_4(AAF)$, then $c \in E^+$. (1)

Proof of Claim. Let D be the characteristic function of $SR_4(AAF)$, thus $E = \bigcup_{i=1, \dots, \infty} D^i(\emptyset)$. The proof is carried out by induction on the value of i . If $b \in D^1(\emptyset)$, then c does not attack b in $SR_4(AAF)$. By Definition 10, it must be either

- $b \rightarrow c$ and $|\mathcal{S}_c| \not\geq |\mathcal{S}_b|$, or
- $b \rightarrow c, c \rightarrow b$ and $|\mathcal{S}_b| > |\mathcal{S}_c|$.

In either case, $c \in E^+$. Assume the claim holds for $i = n$. If $b \in D^{n+1}(\emptyset)$, we distinguish between two cases. If c does not attack b in $SR_4(AAF)$, by the same reasoning as above, $c \in E^+$; otherwise, there must be a $b' \in D^n(\emptyset)$ such that b' attacks c in $SR_4(AAF)$. Applying the IH, we have $c \in E^+$.

Therefore, $b \in E^+$. That is, there is an $a \in E$ such that $a \rightarrow b$. By Definition 10, it must be the case that either a attacks b or b attacks a in $SR_4(AAF)$. In either case, there is an $a' \in E$ such that a' attacks b in $SR_4(AAF)$. Since b is arbitrary, we conclude that E defends c in $SR_4(AAF)$. Thus, $c \in E$, a contradiction!

$\square$

11.4. $SR_3 \times P_4$

Lemma 11.3. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, a set $E \subseteq \mathcal{A}$ and a SCC $S \in SCCS_{AAF}$, for all $a \in S$, we have:

- (1) $a \in UP_{AAF}(S, E)$ iff for all $b \in E \setminus S$, it holds that $b \not\rightarrow a$.

(2) $a \in U_{AAF}(S, E)$ iff for all $b \in E \setminus S$, then $b \not\rightarrow a$ and for all $b \in \mathcal{A} \setminus S$ such that $b \rightarrow a$, there is a $c \in E$ such that $c \rightarrow b$.

Definition 24. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set $C \subseteq \mathcal{A}$, a set $E \subseteq \mathcal{A}$ is an *admissible set in C* iff $E \subseteq C$ and E is admissible in $SR_3(AAF)$. The set of admissible sets in C is denoted as $\mathcal{AS}(AAF, C)$.

Definition 25. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set $C \subseteq \mathcal{A}$, a set $E \subseteq \mathcal{A}$ is a *complete extension in C* iff E is an admissible set in C and every $a \in C$ that is defended by E in $SR_3(AAF)$ belongs to E . The set of complete extensions in C is denoted as $\mathcal{CE}(AAF, C)$.

Definition 26. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set $C \subseteq \mathcal{A}$, a set $E \subseteq \mathcal{A}$ is a *preferred extension in C* if and only if E is a maximal element (w.r.t. set inclusion) of $\mathcal{AS}(AAF, C)$. The set of preferred extensions in C is denoted as $\mathcal{PE}(AAF, C)$.

Definition 27. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set $C \subseteq \mathcal{A}$, a set $E \subseteq \mathcal{A}$ is a *grounded extension in C* if and only if E is the least element (w.r.t. set inclusion) of $\mathcal{CE}(AAF, C)$. The set of grounded extensions in C is denoted as $\mathcal{GE}(AAF, C)$.

According to [11], the grounded extension in C exists and is unique for any $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and any set $C \subseteq \mathcal{A}$.

Definition 28. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set $C \subseteq \mathcal{A}$, a set $E \subseteq \mathcal{A}$ is a *stable extension in C* if and only if $E \subseteq C$ and E is a stable extension in $SR_3(AAF)$. The set of stable extensions in C is denoted as $\mathcal{SE}(AAF, C)$.

Lemma 11.4. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, a set of arguments $E \subseteq \mathcal{A}$ that is admissible in $SR_3(AAF)$, and an argument a defended by E in $SR_3(AAF)$, with $SCC_{AAF}(a)$ denoted as S , it holds that:

- $a \in U_{AAF}(S, E)$; and
- a is defended by $E \cap S$ in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$.

Proof. To prove that $a \in U_{AAF}(S, E)$, we need to show, by Lemma 11.3, that:

- (1) for all $b \in E \setminus S$, it holds that $b \not\rightarrow a$, and
- (2) for all $b \in \mathcal{A} \setminus S$ such that $b \rightarrow a$, there is a $c \in E$ such that $c \rightarrow b$.

For (1), suppose there is a $b \in E \setminus S$ such that $b \rightarrow a$. Then $a \not\rightarrow b$ (otherwise, $b \in S$, a contradiction!). Thus, by Definition 10, b attacks a in $SR_3(AAF)$. Since a is defended by E in $SR_3(AAF)$, there must be a $c \in E$ such that c attacks b in $SR_3(AAF)$. This implies that E is not conflict-free in $SR_3(AAF)$, a contradiction! For (2), let $b \in \mathcal{A} \setminus S$ be such that $b \rightarrow a$. Thus, $a \not\rightarrow b$. By Definition 10, b attacks a in $SR_3(AAF)$. Since a is defended by E in $SR_3(AAF)$, there must be a $c \in E$ such that c attacks b in $SR_3(AAF)$. Therefore, it must be the case that $c \rightarrow b$ by Definition 10.

The second item remains to be considered. We first note that it follows from the first item that $E \cap S \subseteq U_{AAF}(S, E)$ because every element of $E \cap S$ is defended by E in $SR_3(AAF)$. Let $b \in UP_{AAF}(S, E)$ be such that b attacks a in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$. Thus, b attacks a in $SR_3(AAF)$. Since $b \in UP_{AAF}(S, E)$, by Definition 10, for all $c \in E \setminus S$, c does not attack b in $SR_3(AAF)$. But a is defended by E in $SR_3(AAF)$, therefore there must be a $c \in E \cap S$ such that c attacks b in $SR_3(AAF)$ (thus in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$). It follows that a is defended by $E \cap S$ in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$ since b is arbitrary. $\square$

Lemma 11.5. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $E \subseteq \mathcal{A}$ be a set of arguments such that, for all $S \in SCCS_{AAF}$:

$$(E \cap S) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S,E)).$$

Given $\hat{S} \in SCCS_{AAF}$ and an argument $a \in U_{AAF}(\hat{S}, E)$. Then for all $b \in \mathcal{A} \setminus \hat{S}$ such that b attacks a in $SR_3(AAF)$, there is an $a' \in E$ such that a' attacks b in $SR_3(AAF)$.

Proof. By Definition 10, $b \rightarrow a$. Since $a \in U_{AAF}(\hat{S}, E)$, there must be a $c \in E$ such that $c \rightarrow b$. If c attacks b in $SR_3(AAF)$, then we are done. Otherwise, by Definition 10, the only possibility is that $c \rightarrow b$, $b \rightarrow c$, and $|\mathcal{S}_c| < |\mathcal{S}_b|$. In that case, b attacks c in $SR_3(AAF)$. We show that there must be a $c' \in E$ such that c' attacks b in $SR_3(AAF)$. We denote $SCC_{AAF}(b)$ as S' . If $b \in S' \setminus UP_{AAF}(S', E)$, then, by definition, there must be a $c' \in E \setminus S'$ such that $c' \rightarrow b$. Note that $b \not\rightarrow c'$ (otherwise $c' \in S'$). Thus, c' attacks b in $SR_3(AAF)$ by Definition 10. If $b \in UP_{AAF}(S', E)$, then note that $E \cap S' \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S',E)}, U_{AAF}(S',E))$, that $c \in E \cap S'$, and that b attacks c in $SR_3(AAF \downarrow_{UP_{AAF}(S',E)})$. Therefore, there must be a $c' \in E \cap S'$ such that c' attacks b in $SR_3(AAF \downarrow_{UP_{AAF}(S',E)})$ (thus in $SR_3(AAF)$). $\square$

Lemma 11.6. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, let $E \subseteq \mathcal{A}$ be a set of arguments such that, for all $S \in SCCS_{AAF}$:

$$(E \cap S) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S,E)).$$

Given a $\hat{S} \in SCCS_{AAF}$ and an argument $a \in U_{AAF}(\hat{S}, E)$ that is defended by $E \cap \hat{S}$ in $SR_3(AAF \downarrow_{UP_{AAF}(\hat{S},E)})$, then a is defended by E in $SR_3(AAF)$.

Proof. We first note that $E \cap \hat{S} \subseteq UP_{AAF}(\hat{S}, E)$. Let $b \in \mathcal{A}$ with b attacking a in $SR_3(AAF)$. We distinguish between three cases. (1) If $b \in UP_{AAF}(\hat{S}, E)$, then b attacks a in $SR_3(AAF \downarrow_{UP_{AAF}(\hat{S},E)})$. Since a is defended by $E \cap \hat{S}$ in $SR_3(AAF \downarrow_{UP_{AAF}(\hat{S},E)})$, there must be a $c \in E \cap \hat{S}$ such that c attacks b in $SR_3(AAF \downarrow_{UP_{AAF}(\hat{S},E)})$ (and, thus, in $SR_3(AAF)$). (2) If $b \in \hat{S} \setminus UP_{AAF}(\hat{S}, E)$, by definition, there must be a $c \in E \setminus \hat{S}$ such that $c \rightarrow b$. Note that $b \not\rightarrow c$ (otherwise $c \in \hat{S}$). Thus c attacks b in $SR_3(AAF)$ by Definition 10. (3) If $b \notin \hat{S}$, by Lemma 11.5, there is an $a' \in E$ such that a' attacks b in $SR_3(AAF)$. $\square$

Lemma 11.7. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, $\forall C \subseteq \mathcal{A}$, $E \in \mathcal{AS}(AAF, C)$ if and only if $\forall S \in SCCS_{AAF}$:

$$(E \cap S) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S,E) \cap C).$$

Proof. From left to right. We first show that (1) $E \cap S \subseteq U_{AAF}(S, E) \cap C$. It suffices to show that $E \cap S \subseteq U_{AAF}(S, E)$ since $E \subseteq C$. Let $a \in E \cap S$. Since E is admissible in $SR_3(AAF)$ and a is defended by E in $SR_3(AAF)$, it follows from Lemma 11.4 that $a \in U_{AAF}(S, E)$. We then show that (2) $E \cap S$ is admissible in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$. It is easy to see that $E \cap S$ is conflict-free in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$. Let $a \in E \cap S$. Since E is admissible in $SR_3(AAF)$ and a is defended by E in $SR_3(AAF)$, it follows from Lemma 11.4 that a is defended by $E \cap S$ in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$.

From right to left. Since $\forall S \in SCCS_{AAF}$ such that $(E \cap S) \subseteq C$, it follows that (1) $E \subseteq C$. We then show that (2) E is conflict-free in $SR_3(AAF)$. Let $a, b \in E$. If $SCCS_{AAF}(a) = SCCS_{AAF}(b) = S$, since

$E \cap S$ is admissible in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$, there is no attack between them in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$ (thus in $SR_3(AAF)$). If $SCCS_{AAF}(a) \neq SCCS_{AAF}(b)$, since $E \cap SCCS_{AAF}(b) \subseteq U_{AAF}(SCCS_{AAF}(b), E)$ and $a \in E \setminus SCCS_{AAF}(b)$, it follows that $a \not\rightarrow b$. Thus, by Definition 10, a does not attack b in $SR_3(AAF)$. Likewise, b does not attack a in $SR_3(AAF)$. Finally, we show that (3) E can defend itself in $SR_3(AAF)$. Let $a \in E$. Denote $SCCS_{AAF}(a)$ as $\hat{S}$. Since $E \cap \hat{S}$ is admissible in $SR_3(AAF \downarrow_{UP_{AAF}(\hat{S},E)})$ and $a \in E \cap \hat{S}$, it follows that a is defended by $E \cap \hat{S}$ in $SR_3(AAF \downarrow_{UP_{AAF}(\hat{S},E)})$. Thus, applying Lemma 11.6, we have that a is defended by E in $SR_3(AAF)$. $\square$

11.4.1. Complete Semantics

Proposition 44. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubseteq \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, $\forall C \subseteq \mathcal{A}$, $E \in \mathcal{CE}(AAF, C)$ if and only if $\forall S \in SCCS_{AAF}$:

$$(E \cap S) \in \mathcal{CE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C).$$

Proof. From left to right. It follows from Lemma 11.7 that $(E \cap S) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$. It remains to be shown that $\forall a \in U_{AAF}(S, E) \cap C$ such that a is defended by $E \cap S$ in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$, it must be the case that $a \in E \cap S$. Since a is defended by $E \cap S$ in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$, by Lemma 11.7, a is defended by E in $SR_3(AAF)$. Since $a \in C$ and E is a complete extension in C , $a \in E$.

From right to left. It follows from Lemma 11.7 that $E \in \mathcal{AS}(AAF, C)$. It remains to be shown that $\forall a \in C$, if a is defended by E in $SR_3(AAF)$, then $a \in E$. By Lemma 11.7, we know that E is admissible in $SR_3(AAF)$. Since a is defended by E in $SR_3(AAF)$, by Lemma 11.4, $a \in U_{AAF}(S, E)$ and a is defended by $E \cap S$ in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$. Thus $a \in E \cap S$ since $(E \cap S) \in \mathcal{CE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$. $\square$

11.4.2. Preferred Semantics

Lemma 11.8. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubseteq \rangle$, an admissible set E in $SR_3(AAF)$, and an $S \in SCCS_{AAF}$, let $\hat{E}$ be a set of arguments such that:

- $E \cap S \subseteq \hat{E} \subseteq U_{AAF}(S, E)$;
- $\hat{E}$ is admissible in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$.

It holds that $E \cup \hat{E}$ is admissible in $SR_3(AAF)$.

Proof. We first show that $E \cup \hat{E}$ is conflict-free in $SR_3(AAF)$. Suppose not. Then the only possibility is that there is an $a \in E \setminus \hat{E}$ (thus $a \in E \setminus S$) and a $b \in \hat{E} \setminus E$ such that a attacks b or b attacks a in $SR_3(AAF)$. In the former case, by Definition 10, it must be that $a \rightarrow b$. But that contradicts $b \in \hat{E} \subseteq U_{AAF}(S, E)$. In the latter case, since E is admissible in $SR_3(AAF)$, there must be an $a' \in E$ such that a' attacks b in $SR_3(AAF)$. Moreover, $a' \notin \hat{E}$ (otherwise $\hat{E}$ is not conflict-free in $SR_3(AAF)$), thus $a' \in E \setminus \hat{E}$. This is also impossible for the same reason as in the former case.

We then show $E \cup \hat{E}$ can defend itself in $SR_3(AAF)$. It suffices to show that for all $a \in \hat{E} \setminus E$, it is the case that $E \cup \hat{E}$ defends a in $SR_3(AAF)$. Let $b \in \mathcal{A}$ be such that b attacks a in $SR_3(AAF)$. We distinguish between three cases. (a) If $b \in U_{AAF}(S, E)$, then b attacks a in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$. Since $\hat{E}$ is admissible in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$, there must be a $c \in \hat{E}$ such that c attacks b in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$ (thus in $SR_3(AAF)$). (b) If $b \in S \setminus U_{AAF}(S, E)$, by definition, there must be a $c \in E \setminus S$ such that $c \rightarrow b$. Since $b \not\rightarrow c$, by Definition 10, c attacks b in $SR_3(AAF)$. (c) If $b \in \mathcal{A} \setminus S$. We first note that, by Lemma 11.7, we have (1) $\forall S \in SCCS_{AAF}$: $(E \cap S) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E))$. We also have (2) $a \in U_{AAF}(S, E)$.

Based on (1) and (2), we can apply Lemma 11.5 and obtain that there is a $c \in E$ such that c attacks b in $SR_3(AAF)$. $\square$

Proposition 45. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, $\forall C \subseteq \mathcal{A}$, $E \in \mathcal{PE}(AAF, C)$ if and only if $\forall S \in SCCS_{AAF}$:

$$(E \cap S) \in \mathcal{PE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C).$$

Proof. From left to right. By Lemma 11.7, it follows that $(E \cap S) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$. Suppose, toward a contradiction, that $E \cap S$ is not a maximal element (w.r.t set inclusion) in $\mathcal{AS}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$, i.e., there is an $\hat{E} \in \mathcal{CE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$ such that $\hat{E} \supset (E \cap S)$. Note that:

- $E \cap S \subseteq \hat{E} \subseteq U_{AAF}(S, E)$;
- $\hat{E}$ is admissible in $SR_3(AAF \downarrow_{UP_{AAF}(S,E)})$.

Thus we can apply Lemma 11.8 and obtain that $E \cup \hat{E}$ is admissible in $SR_3(AAF)$. Since $E \cup \hat{E} \supset E$, that contradicts $E \in \mathcal{PE}(AAF, C)$.

From right to left. By Lemma 11.7, it follows that $E \in \mathcal{AS}(AAF, C)$. Suppose, toward a contradiction, that there is an $\hat{E} \supset E$ and $\hat{E} \in \mathcal{AS}(AAF, C)$. Then there is an $S \in SCCS_{AAF}$ such that $\hat{E} \cap S \supset E \cap S$. Let $\hat{S} \in SCCS_{AAF}$ be such that

- (1) $\hat{E} \cap \hat{S} \supset E \cap \hat{S}$,
- (2) for any $S \in SCCS_{AAF}$ such that S is an ancestor of $\hat{S}$ in the condensation of the graph $(\mathcal{A}, \rightarrow)$, $\hat{E} \cap S = E \cap S$.

By Lemma 11.7, we have that

$$(\hat{E} \cap \hat{S}) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(\hat{S}, \hat{E})}, U_{AAF}(\hat{S}, \hat{E}) \cap C).$$

But, by (2), it is easy to see that $UP_{AAF}(\hat{S}, \hat{E}) = UP_{AAF}(\hat{S}, E)$ and $U_{AAF}(\hat{S}, \hat{E}) = U_{AAF}(\hat{S}, E)$. Thus

$$(\hat{E} \cap \hat{S}) \in \mathcal{AS}(AAF \downarrow_{UP_{AAF}(\hat{S}, E)}, U_{AAF}(\hat{S}, E) \cap C).$$

But $\hat{E} \cap \hat{S} \supset E \cap \hat{S}$ contradicts $(E \cap \hat{S}) \in \mathcal{PE}(AAF \downarrow_{UP_{AAF}(\hat{S}, E)}, U_{AAF}(\hat{S}, E) \cap C)$. $\square$

11.4.3. Grounded Semantics

Proposition 46. Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, $\forall C \subseteq \mathcal{A}$, $E \in \mathcal{GE}(AAF, C)$ if and only if $\forall S \in SCCS_{AAF}$:

$$(E \cap S) \in \mathcal{GE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C).$$

Proof. From left to right. By Proposition 44, we have that $\forall S \in SCCS_{AAF}$: $(E \cap S) \in \mathcal{CE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$. Suppose, toward a contradiction, that there is an $S \in SCCS_{AAF}$ such that $E \cap S$ is not the least element in $\mathcal{CE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$. Let $\hat{S} \in SCCS_{AAF}$ be such that:

- for any $S \in SCCS_{AAF}$ such that S is an ancestor of $\hat{S}$ in the condensation of the graph $(\mathcal{A}, \rightarrow)$: $E \cap S \in \mathcal{GE}(AAF \downarrow_{UP_{AAF}(S,E)}, U_{AAF}(S, E) \cap C)$.

- $\exists \hat{E} \subset (E \cap \hat{S}) : \hat{E} \in \mathcal{GE}(AAF \downarrow_{UP_{AAF}(\hat{S}, E)}, U_{AAF}(\hat{S}, E) \cap C)$

We can thus construct a set $E' \subseteq \mathcal{A}$ such that:

- for any $S \in SCCS_{AAF}$ such that S is an ancestor of $\hat{S}$ in the condensation of the graph $(\mathcal{A}, \rightarrow)$:
 $(E' \cap S) = (E \cap S)$;
- $E' \cap \hat{S} = \hat{E}$;
- $\forall S \in SCCS_{AAF} : (E' \cap S) \in \mathcal{GE}(AAF \downarrow_{UP_{AAF}(S, E')}, U_{AAF}(S, E') \cap C)$.

It is easy to see that $E' \in \mathcal{CE}(AAF, C)$ by Proposition 44. But $E \not\subseteq E'$, a contradiction!

From right to left. By Proposition 44, we have $E \in \mathcal{CE}(AAF, C)$. Suppose that $\exists E' \in \mathcal{GE}(AAF, C) : E' \subset E$. Then there must be an $S \in SCCS_{AAF}$ such that $E' \cap S \subset E \cap S$. Let $\hat{S} \in SCCS_{AAF}$ be such that:

- for any $S \in SCCS_{AAF}$ such that S is an ancestor of $\hat{S}$ in the condensation of the graph $(\mathcal{A}, \rightarrow)$:
 $E' \cap S = E \cap S$;
- $E' \cap \hat{S} \subset E \cap \hat{S}$.

Since $UP_{AAF}(\hat{S}, E') = UP_{AAF}(\hat{S}, E)$, and $U_{AAF}(\hat{S}, E') = U_{AAF}(\hat{S}, E)$, and $E' \cap \hat{S} \in \mathcal{CE}(AAF \downarrow_{UP_{AAF}(\hat{S}, E')}, U_{AAF}(\hat{S}, E') \cap C)$ (by Proposition 44), we have $E' \cap \hat{S} \in \mathcal{CE}(AAF \downarrow_{UP_{AAF}(\hat{S}, E)}, U_{AAF}(\hat{S}, E) \cap C)$. But $E' \cap \hat{S} \subset E \cap \hat{S}$, a contradiction! $\square$

11.4.4. Stable Semantics

Lemma 11.9. *Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, E is a stable extension of $SR_3(AAF)$ if and only if $\forall S \in SCCS_{AAF}$, it is the case that $E \cap S$ is a stable extension of $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$.*

Proof. From left to right. We need to show that

- (1) $E \cap S \subseteq UP_{AAF}(S, E)$;
- (2) $E \cap S$ is conflict-free in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$;
- (3) $\forall a \in UP_{AAF}(S, E) : a \notin (E \cap S)$ implies that $\exists b \in E \cap S$ such that b attacks a in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$.

For (1), let $a \in E \cap S$ be arbitrary. Suppose, toward a contradiction, that there is a $b \in E \setminus S$ such that $b \rightarrow a$. We have $a \not\rightarrow b$ since otherwise $b \in S$. Thus b attacks a in $SR_3(AAF)$ by Definition 10, contradicting that E is conflict-free in $SR_3(AAF)$. For (2), it is easy to see that $E \cap S$ is conflict-free in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$ since otherwise E is not conflict-free in $SR_3(AAF)$. For (3), let $a \in UP_{AAF}(S, E) \setminus (E \cap S)$ be arbitrary. Since $a \in UP_{AAF}(S, E)$, there is no $b \in E \setminus S$ such that $b \rightarrow a$. Thus, by Definition 10, there is no $b \in E \setminus S$ such that b attacks a in $SR_3(AAF)$. But E is a stable extension of $SR_3(AAF)$, thus it can only be that $\exists b \in E \cap S$ such that b attacks a in $SR_3(AAF)$ (thus in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$).

From right to left. By Lemma 11.7, we have that E is admissible in $SR_3(AAF)$, thus E is conflict-free in $SR_3(AAF)$. Let $a \in \mathcal{A} \setminus E$ be arbitrary. We denote $SCCS_{AAF}(a)$ as S . If $b \notin UP_{AAF}(S, E)$, then $\exists b \in E \setminus S : b \rightarrow a$. We have $a \not\rightarrow b$, since otherwise $b \in S$. Thus, by Definition 10, b attacks a in $SR_3(AAF)$. If $b \in UP_{AAF}(S, E)$, since $E \cap S$ is a stable extension of $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$, there must be a $b \in E \cap S$ such that b attacks a in $SR_3(AAF \downarrow_{UP_{AAF}(S, E)})$ (thus in $SR_3(AAF)$). $\square$

Proposition 47. *Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, $\forall C \subseteq \mathcal{A}$, $E \in \mathcal{SE}(AAF, C)$ if and only if $\forall S \in SCCS_{AAF}$:*

$$(E \cap S) \in \mathcal{SE}(AAF \downarrow_{UP_{AAF}(S, E)}, U_{AAF}(S, E) \cap C).$$

Proof. From left to right. By Lemma 11.9, $E \cap S$ is a stable extension of $AAF \downarrow_{UP_{AAF}(S,E)}$. Thus it suffices to show that $E \cap S \subseteq U_{AAF}(S, E)$. Let $a \in E \cap S$ be arbitrary and let $b \in \mathcal{A} \setminus S$ be such that $b \rightarrow a$. We have $a \not\rightarrow a$, since otherwise $b \in S$. Thus, by Definition 10, b attacks a in $SR_3(AAF)$. Since E is conflict-free, $b \notin E$. Thus, E attacks b in $SR_3(AAF)$. Therefore, by Definition 10, there must be a $c \in E$ such that $c \rightarrow b$. Since b is arbitrary, $a \in U_{AAF}(S, E)$.

From right to left. By Lemma 11.9, we have that E is a stable extension of $SR_3(AAF)$. Besides, it is easy to see that $E \subseteq C$. $\square$

Proposition 48 ($SR_3 \times P_4$). All the four kinds of Dung semantics for SR_3 satisfy Principle 4.

Proof. Let us consider, for example, the stable semantics for SR_3 . By definition, it is easy to see that E is a stable extension of $SR_3(AAF)$ if and only if $E \in \mathcal{SE}(AAF, \mathcal{A})$ for any given $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and $E \subseteq \mathcal{A}$. For any AAF with $|SCCS_{AAF}| = 1$, we define the function $\mathcal{B}$ such that $\mathcal{B}(AAF, C) = \mathcal{SE}(AAF, C)$. Then, by Proposition 47, we have that the function $\mathcal{G} = \mathcal{SE}$ satisfies all the conditions in Principle 4.

As far as complete, preferred and grounded semantics for SR_3 are concerned, the proofs are similar and are based on Propositions 44, 45 and 46 respectively. $\square$

11.5. $AR_1^F - AR_4^F$

Proposition 49 ($AR_1^F \times P_1, P_2$). None of the four kinds of Dung semantics for AR_1^F satisfy Principle 1. Thus, they do not satisfy Principle 2 either.

Proof. A counterexample is Figure 8. $\square$

Proposition 50 ($AR_2^F \times P_1$). All four kinds of Dung semantics for AR_2^F satisfy Principle 1.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let $a, b \in \mathcal{A}$. It suffices to show that if $a \rightarrow b$ then there must be an attack between a and b in $AR_2^F(AAF) = \bigcup_{\alpha \in \mathcal{S}} PR_2(AAP(AAF, \alpha))$. Assume $a \rightarrow b$ and let $\alpha \in \mathcal{S}$. (1) If $b \sqsubset \alpha$ and $a \not\sqsubset \alpha$, then $b \succ a$. We consider two cases. If $b \rightarrow a$, then it must be that b attacks a in $PR_2(AAP(AAF, \alpha))$. If $b \not\rightarrow a$ in AAF , since $b \succ a$, it must also be that b attacks a in $PR_2(AAP(AAF, \alpha))$. (2) Otherwise $b \not\succ a$. Thus a attacks b in $PR_2(AAP(AAF, \alpha))$. $\square$

Proposition 51 ($AR_3^F \times P_1$). All four kinds of Dung semantics for AR_3^F satisfy Principle 1.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let $a, b \in \mathcal{A}$. It suffices to show that if $a \rightarrow b$ then there must be an attack between a and b in $AR_3^F(AAF) = \bigcup_{\alpha \in \mathcal{S}} PR_3(AAP(AAF, \alpha))$. Assume $a \rightarrow b$ and let $\alpha \in \mathcal{S}$. (1) If $b \not\rightarrow a$, then a attacks b in $PR_3(AAP(AAF, \alpha))$. (2) If $b \rightarrow a$, since it must be either $b \not\succ a$ or $a \not\succ b$, there must be an attack between them in $PR_3(AAP(AAF, \alpha))$. $\square$

Proposition 52 ($AR_4^F \times P_1$). All four kinds of Dung semantics for AR_4^F satisfy Principle 1.

Proof. Obviously from Proposition 51. $\square$

Proposition 53 ($AR_2^F \times P_2$). None of the four kinds of Dung semantics for AR_2^F satisfy Principle 2.

Proof. A counterexample is Figure 8. $\square$

Proposition 54 ($AR_3^F \times P_2$). All four kinds of Dung semantics for AR_3^F satisfy Principle 2.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given. It suffices to show that any admissible set $E \subseteq \mathcal{A}$ in $AR_3^F(AAF)$ is also admissible in $\langle \mathcal{A}, \rightarrow \rangle$. Suppose, toward a contradiction, that E is not admissible in $\langle \mathcal{A}, \rightarrow \rangle$. By Proposition 51, it must be the case that there is a $c \in E$ such that E cannot defend c in $\langle \mathcal{A}, \rightarrow \rangle$. That is, there is a $b \in \mathcal{A}$ such that $b \rightarrow c$ and $a \not\rightarrow b$ for all $a \in E$. Thus, in particular, $c \not\rightarrow b$. By Definition 10, it follows that b attacks c in $AR_3^F(AAF)$. Since $a \not\rightarrow b$ for all $a \in E$, it holds that a does not attack b for all $a \in E$ in $AR_3^F(AAF)$. Thus, E cannot defend c in $AR_3^F(AAF)$, a contradiction! $\square$

Proposition 55 ($AR_4^F \times P_2$). The complete semantics, preferred semantics and stable semantics for AR_4^F do not satisfy Principle 2, whereas the grounded semantics for AR_4^F does satisfy Principle 2.

Proof. For the first half, a counterexample is Figure 20. Note that $\{b\}$ is a complete, preferred and stable extension of $AR_4^F(AAF)$, but it is not admissible in $\langle \mathcal{A}, \rightarrow \rangle$.

For the second half of the lemma, let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let $E \subseteq \mathcal{A}$ be the grounded extension of $AR_4^F(AAF)$. Since E is conflict-free in $\langle \mathcal{A}, \rightarrow \rangle$ by Proposition 52, we only need to show that E can defend all its members in $\langle \mathcal{A}, \rightarrow \rangle$. Let D be the characteristic function of $AR_4^F(AAF)$, thus $E = \bigcup_{i=1, \dots, \infty} D^i(\emptyset)$. We prove, by induction on the value of i , that E can defend $D^i(\emptyset)$ in $\langle \mathcal{A}, \rightarrow \rangle$ for $i = 1, \dots, \infty$. If $i = 1$, let $c \in D^1(\emptyset) = D(\emptyset)$ be arbitrary. For any argument b such that $b \rightarrow c$ and $c \not\rightarrow b$, it must be that b attacks c in $AR_4^F(AAF)$ by Definition 10. Thus such a b does not exist, so E can defend $D^1(\emptyset)$ in $\langle \mathcal{A}, \rightarrow \rangle$. Now suppose that E can defend $D^n(\emptyset)$ in $\langle \mathcal{A}, \rightarrow \rangle$. Let $c \in D^{n+1}(\emptyset)$ be arbitrary. For any argument b such that $b \rightarrow c$ and $c \not\rightarrow b$, it must be that b attacks c in $AR_4^F(AAF)$ by Definition 10. Since $c \in D^{n+1}(\emptyset) = D(D^n(\emptyset))$, there must be an $a \in D^n(\emptyset)$ such that a attacks b in $AR_4^F(AAF)$. If $a \not\rightarrow b$, it must hold that $b \rightarrow a$ by Definition 10. By IH, there is an $a' \in E$ such that $a' \rightarrow b$, as desired. $\square$

Proposition 56 ($AR_1^F, AR_3^F \times P_3$). The grounded, complete, preferred semantics for AR_1^F and AR_3^F satisfy Principle 3, whereas the stable semantics for AR_1^F and AR_3^F do not satisfy Principle 3.

Proof. For any $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$, if a set of arguments $\mathcal{U} \subseteq \mathcal{A}$ is unattacked in $\langle \mathcal{A}, \rightarrow \rangle$, then it is also unattacked in $AR_1^F(AAF)$ and $AR_3^F(AAF)$. Thus the first half follows from [10] directly.

For the second half of the lemma, consider the following $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S} = \{\alpha\}, \sqsubset \rangle$ in Figure 10. It is easy to see that $AR_1^F(AAF) = AR_3^F(AAF) = \langle \mathcal{A}, \rightarrow \rangle$. In particular, $\{a\}$ is unattacked and is a stable extension of $AAF \downarrow_{\{a\}}$. But there is no stable extension of $\langle \mathcal{A}, \rightarrow \rangle$. $\square$

Proposition 57 ($AR_2^F, AR_4^F \times P_3$). None of the four kinds of Dung semantics for AR_2^F and/ or AR_4^F satisfy Principle 3.

Proof. The counterexamples are the same as in Proposition 38. $\square$

Proposition 58 ($AR_1^F, AR_2^F, AR_4^F \times P_4$). None of the four kinds of Dung semantics for AR_1^F, AR_2^F , and AR_4^F satisfy Principle 4.

Proof. Similarly to the proof of Proposition 40. $\square$

Proposition 59 ($AR_3^F \times P_4$). All four kinds of Dung semantics for AR_3^F satisfy Principle 4.

Proof. The proofs are similar to those in Section 11.4. We just substitute AR_3^F for SR_3 . $\square$

Proposition 60 ($AR_1^F \times P_5$). The grounded, complete, and preferred semantics for AR_1^F do not satisfy Principle 5, whereas the stable semantics for AR_1^F does satisfy Principle 5.

Proof. Similarly to the proof of Proposition 7. $\square$

Proposition 61 ($AR_2^F \times P_5$). None of the four kinds of Dung semantics for AR_2^F satisfy Principle 5.

Proof. Similarly to the proof of Proposition 41. $\square$

Proposition 62 ($AR_3^F \times P_5$). All four kinds of Dung semantics for AR_3^F satisfy Principle 5.

Proof. The proof can be obtained by substituting AR_3^F for SR_3 in the proof of Proposition 42. $\square$

Proposition 63 ($AR_4^F \times P_5$). The complete, preferred and stable semantics for AR_4^F does not satisfy Principle 5, whereas the grounded semantics for AR_4^F does satisfy Principle 5.

Proof. The proof is similar to that of Proposition 43. $\square$

11.6. $AR_1^A - AR_4^A$

Proposition 64 ($AR_1^A \times P_1, P_2$). None of the four kinds of Dung semantics for AR_1^A satisfies P_1 . Thus, none of them satisfies P_2 .

Proof. See Table 3. $\square$

Proposition 65 ($AR_2^A \times P_1$). All four kinds of Dung semantics for AR_2^A satisfy principle P_1 .

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \square \rangle$ be given and let $a, b \in \mathcal{A}$. It suffices to show that, for all α , if $a \rightarrow b$ then there must be an attack between a and b in $PR_2(AAP(AAF, \alpha))$. This can be shown in the same way as in the proof of Prop. 50. $\square$

Proposition 66 ($AR_3^A \times P_1$). All four kinds of Dung semantics for AR_3^A satisfy principle P_1 .

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \square \rangle$ be given and let $a, b \in \mathcal{A}$. It suffices to show that, for all α , if $a \rightarrow b$ then there must be an attack between a and b in $PR_3(AAP(AAF, \alpha))$. This can be shown in the same way as in the proof of Prop. 51. $\square$

Proposition 67 ($AR_4^A \times P_1$). All four kinds of Dung semantics for AR_4^A satisfy principle P_1 .

Proof. Obviously from Prop. 66. $\square$

Proposition 68 ($AR_2^A \times P_2$). None of the four kinds of Dung semantics for AR_2^A satisfy P_2 .

Proof. A counterexample is provided in Figure 8. $\square$

Proposition 69 ($AR_3^A \times P_2$). All four kinds of Dung semantics for AR_3^A satisfy Principle P_2 .

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given. It suffices to show that, for all agents $\alpha \in \mathcal{S}$, any set $E \subseteq \mathcal{A}$ that is admissible in $PR_3(AAP(AAF, \alpha))$ is also admissible in $\langle \mathcal{A}, \rightarrow \rangle$. This can be shown in the same way as in the proof of Prop. 54. $\square$

Proposition 70 ($AR_4^A \times P_2$). The complete, preferred, and stable semantics for AR_4^A do not satisfy Principle 2, whereas the grounded semantics for AR_4^A does satisfy Principle 2.

Proof. The same as the proof of Prop. 55. $\square$

Proposition 71 ($AR_1^A, AR_3^A \times P_3$). The grounded, complete, preferred semantics for AR_1^A and AR_3^A satisfy Principle 3, whereas the stable semantics for AR_1^A and AR_3^A do not satisfy Principle 3.

Proof. For any $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and agent $\alpha \in \mathcal{S}$, if a set of arguments $\mathcal{U} \subseteq \mathcal{A}$ is unattacked in $\langle \mathcal{A}, \rightarrow \rangle$, then it is also unattacked in $PR_1(AAP(AAF, \alpha))$ and $PR_3(AAP(AAF, \alpha))$. Thus the first half follows from [10] directly.

For the second half of the lemma, a counterexample is provided in Figure 10. $\square$

Proposition 72 ($AR_2^A, AR_4^A \times P_3$). None of the four kinds of Dung semantics for AR_2^A and/or AR_4^A satisfy Principle 3.

Proof. The counterexamples are the same as in Proposition 38. $\square$

Proposition 73 ($AR_1^A, AR_2^A, AR_4^A \times P_4$). None of the four kinds of Dung semantics for AR_1^A, AR_2^A, AR_4^A satisfy Principle 4.

Proof. Similarly to the proof of Proposition 40. $\square$

Proposition 74 ($AR_1^A \times P_5$). None of the grounded, complete, and preferred semantics for AR_1^A satisfy Principle 5, whereas the stable semantics for AR_1^A does satisfy Principle 5.

Proof. See the proof of Prop. 7. $\square$

Proposition 75 ($AR_2^A \times P_5$). None of the four kinds of Dung semantics for AR_2^A satisfies Principle 5.

Proof. Similarly to the proof of Proposition 41. $\square$

Proposition 76 ($AR_3^A \times P_5$). None of the grounded, complete, and preferred semantics for AR_3^A satisfy Principle 5, whereas the stable semantics for AR_3^A does satisfy Principle 5.

Proof. For the first part, consider $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ in Figure 25:

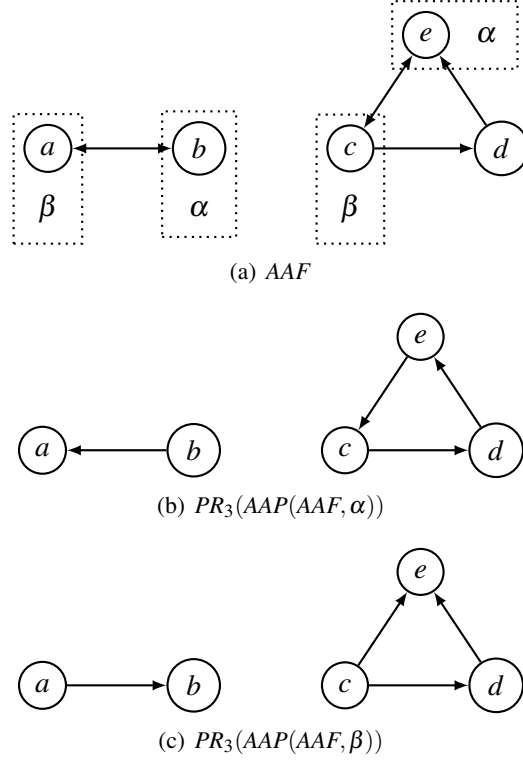


Fig. 25. A counterexample for Proposition 76.

It can be seen that $E = \{b\}$ is the grounded, complete, and preferred extension of $PR_3(AAP(AAF, \alpha))$ and $E' = \{c\}$ is the grounded, complete, and preferred extension of $PR_3(AAP(AAF^E, \beta))$. However, $E \cup E' = \{b, c\}$ is admissible in neither $PR_3(AAP(AAF, \alpha))$ nor $PR_3(AAP(AAF, \beta))$.

For the second part, note that E^* is empty if E is a stable extension of $PR_3(AAP(AAF, \alpha))$ for some $\alpha \in \mathcal{S}$. $\square$

11.7. OR and NBR

Proposition 77 ($OR \times P_1$). All four kinds of Dung semantics for OR satisfy P_1 .

Proof. Trivial. $\square$

Proposition 78 ($OR \times P_2$). None of the four kinds of Dung semantics for OR satisfy P_2 .

Proof. A counterexample is Figure 8. $\square$

Proposition 79 ($NBR \times P_1, P_2$). None of the four kinds of Dung semantics for NBR satisfy P_1 and P_2 .

Proof. A counterexample is Figure 8. $\square$

Proposition 80 ($OR \times P_3$). The complete, grounded and preferred semantics for OR satisfy Principle 3, whereas the stable semantics for OR does not satisfy Principle 3.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let $\mathcal{U} \subseteq \mathcal{A}$ be an unattacked set. Since the set of arguments in $OR(AAF \downarrow_{\mathcal{U}})$ is unattacked in $OR(AAF)$, the first half of the lemma follows directly from [10]. For the second half of the lemma, a counterexample is provided in Figure 26. It is easy to see that $OR(AAF) = \langle \mathcal{A}, \rightarrow \rangle$. In particular, $\{a\}$ is unattacked and is a stable extension of $AAF \downarrow_{\{a\}}$. But there is no stable extension E of $\langle \mathcal{A}, \rightarrow \rangle$ such that $a \in E$. $\square$

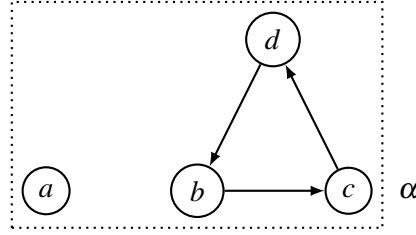


Fig. 26. Stable semantics for OR do not satisfy Principle 3

Proposition 81 ($NBR \times P_3$). The complete, grounded and preferred semantics for NBR satisfy Principle 3, whereas the stable semantics for NBR does not satisfy Principle 3.

Proof. The proof is similar to that of Proposition 80. $\square$

Proposition 82 ($NBR \times P_4$). None of the four kinds of Dung semantics for NBR satisfy Principle 4.

Proof. A counterexample is Figure 22. $\square$

Proposition 83 ($OR \times P_5$). All four kinds of Dung semantics for OR satisfy Principle 5.

Proof. Let an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let $OR(AAF) = (\mathcal{A}', \rightarrow')$. We show that, for any $E \subseteq \mathcal{A}'$, it holds that $OR(AAF^E) = (OR(AAF))^E$, where $(OR(AAF))^E$ is the E -reduct of $OR(AAF)$ defined in [12]. Let $(OR(AAF))^E = (\mathcal{A}'', \rightarrow'')$. We first show that $\mathcal{A}' = \mathcal{A}''$.

$$\begin{aligned}
 & a \in \mathcal{A}' \\
 \text{iff } & a \notin E, a \text{ is not attacked by } E \text{ in } AAF, \text{ and there is an } \alpha \in \mathcal{S} \text{ s.t. } a \sqsubset \alpha && \text{(by definition)} \\
 \text{iff } & \text{there is an } \alpha \in \mathcal{S} \text{ s.t. } a \sqsubset \alpha, a \notin E, \text{ and } a \text{ is not attacked by } E \text{ in } OR(AAF) && \text{(since } E \subseteq \mathcal{A}') \\
 \text{iff } & a \in \mathcal{A}''
 \end{aligned}$$

We then show that $\rightarrow' = \rightarrow''$. For any $a, b \in \mathcal{A}' = \mathcal{A}''$:

$$a \rightarrow' b \text{ iff } a \rightarrow b \text{ iff } a \rightarrow'' b.$$

Thus, $OR(AAF^E) = (OR(AAF))^E$. Therefore, the lemma follows directly from [12]. $\square$

Proposition 84 ($\text{NBR} \times \text{P}_5$). The grounded, complete, and preferred semantics for NBR do not satisfy Principle 5, whereas the stable semantics for NBR do satisfy Principle 5.

Proof. Consider Figure 12. It is easy to see that $\{a\}$ is the grounded extension of $\text{SR}_1(\text{AAF})$ as well as being a complete extension and a preferred extension of it. Let $E = \{a\}$, then AAF^E consists of the single point b . It is also easy to see that $\{b\}$ is the grounded, complete and preferred extension of AAF^E . However, $\{a, b\}$ is not admissible in $\text{NBR}(\text{AAF})$.

For the second half of the proposition, let $\text{AAF} = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given and let E be a stable extension of $\text{NBR}_1(\text{AAF})$. It is easy to know that E^* is empty, thus E' is also empty. $\square$

11.8. $\text{OR} \times \text{P}_4$

11.8.1. Complete, Grounded and Preferred Semantics

In this subsection, let an $\text{AAF}_1 = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ be given. We define $\text{AAF}_2 = \langle \mathcal{A}, \rightarrow, \{\alpha\}, \sqsubset' \rangle$ so that $a \sqsubset' \alpha$ if and only if $\exists \beta \in \mathcal{S}$ such that $a \sqsubset \beta$. It is easy to see that $\text{OR}(\text{AAF}_1) = \text{OR}(\text{AAF}_2)$. We denote $\text{OR}(\text{AAF}_1) = \text{OR}(\text{AAF}_2)$ as $\langle \mathcal{A}', \rightarrow' \rangle$. We also note that $\text{SCCS}_{\text{AAF}_1} = \text{SCCS}_{\text{AAF}_2}$, $\text{UP}_{\text{AAF}_1}(S, E) = \text{UP}_{\text{AAF}_2}(S, E)$ and $\text{U}_{\text{AAF}_1}(S, E) = \text{U}_{\text{AAF}_2}(S, E)$ for any $E \subseteq \mathcal{A}$ and $S \in \text{SCCS}_{\text{AAF}_1}$.

Lemma 11.10. *The following hold:*

- (1) *For any arguments a, b in $\mathcal{A}'$, it is the case that $a \rightarrow' b$ if and only if a attacks b in $\text{SR}_3(\text{AAF}_2)$.*
- (2) *$\mathcal{A}'$ is unattacked in $\text{SR}_3(\text{AAF}_2)$.*

Proof. Immediately from Definition 10. $\square$

Lemma 11.11. *For any $E \subseteq \mathcal{A}$, E is a complete extension in $\text{OR}(\text{AAF}_1) = \text{OR}(\text{AAF}_2)$ if and only if $E \in \mathcal{CE}(\text{AAF}_2, \mathcal{A}')$ where the function $\mathcal{CE}$ is defined in Definition 25.*

Proof. From left to right. We need to show that:

- (1) $E \subseteq \mathcal{A}'$;
- (2) E is conflict-free in $\text{SR}_3(\text{AAF}_2)$;
- (3) for every $a \in E$, E defends a in $\text{SR}_3(\text{AAF}_2)$;
- (4) for every $a \in \mathcal{A}'$, if E defends a in $\text{SR}_3(\text{AAF}_2)$ then $a \in E$.

The first item is trivial. The second item follows directly from Lemma 11.10(1). For (3), let $b \in \mathcal{A}$ be such that b attacks a in $\text{SR}_3(\text{AAF}_2)$. Since $\mathcal{A}'$ is unattacked in $\text{SR}_3(\text{AAF}_2)$ and $a \in E \subseteq \mathcal{A}'$, we have $b \in \mathcal{A}'$. Thus $b \rightarrow' a$ by Lemma 11.10(1). Since E is a complete extension in $\text{OR}(\text{AAF}_2)$, there must be a $c \in E$ such that $c \rightarrow' b$. Thus, by Lemma 11.10(1) again, c attacks b in $\text{SR}_3(\text{AAF}_2)$. Since b is arbitrary, E defends a in $\text{SR}_3(\text{AAF}_2)$. For (4), we show that E defends a in $\text{OR}(\text{AAF}_2)$, thus $a \in E$ since E is a complete extension in $\text{OR}(\text{AAF}_2)$. Let $b \in \mathcal{A}'$ be such that $b \rightarrow' a$, then b attacks a in $\text{SR}_3(\text{AAF}_2)$ by Lemma 11.10(1). Since E defends a in $\text{SR}_3(\text{AAF}_2)$, there must be a $c \in E$ such that c attacks b in $\text{SR}_3(\text{AAF}_2)$. By Lemma 11.10(1) again, $c \rightarrow' b$. Since b is arbitrary, E defends a in $\text{OR}(\text{AAF}_2)$.

From right to left. We need to show that:

- (1) $E \subseteq \mathcal{A}'$;
- (2) E is conflict-free in $\text{OR}(\text{AAF}_2)$;
- (3) for every $a \in E$, E defends a in $\text{OR}(\text{AAF}_2)$;

(4) for every $a \in \mathcal{A}'$, if E defends a in $OR(AAF_2)$ then $a \in E$.

The first item is trivial. The second item follows directly from Lemma 11.10(1). The remaining proof is similar to that for the direction from left to right. $\square$

Lemma 11.12. *The following hold for any $E \subseteq \mathcal{A}$:*

- (1) E is a preferred extension in $OR(AAF_1) = OR(AAF_2)$ if and only if $E \in \mathcal{PE}(AA\mathcal{F}_2, \mathcal{A}')$ where the function $\mathcal{PE}$ is defined in Definition 26.
- (2) E is a grounded extension in $OR(AAF_1) = OR(AAF_2)$ if and only if $E \in \mathcal{GE}(AA\mathcal{F}_2, \mathcal{A}')$ where the function $\mathcal{GE}$ is defined in Definition 27.

Proof. Directly from Lemma 11.11. $\square$

Proposition 85. The complete, preferred and grounded semantics for OR satisfy Principle 4.

Proof. Let us consider, for example, the complete semantics for OR. Let $\mathcal{G}(AA\mathcal{F}_1 \downarrow_X, Y) = \mathcal{CE}(AA\mathcal{F}_2 \downarrow_X, Y \cap \mathcal{A}')$ for any $Y \subseteq X \subseteq \mathcal{A}$. Then $\mathcal{G}(AA\mathcal{F}_1, \mathcal{A}) = \mathcal{G}(AA\mathcal{F}_1 \downarrow_{\mathcal{A}}, \mathcal{A}) = \mathcal{CE}(AA\mathcal{F}_2 \downarrow_{\mathcal{A}}, \mathcal{A} \cap \mathcal{A}') = \mathcal{CE}(AA\mathcal{F}_2, \mathcal{A}')$. By Lemma 11.11, we know that, for any $E \subseteq \mathcal{A}$, E is a complete extension in $OR(AAF_1)$ if and only if $E \in \mathcal{G}(AA\mathcal{F}_1, \mathcal{A})$. It remains to be shown that for any $E \subseteq \mathcal{A}$ and $C \subseteq \mathcal{A}$: $E \in \mathcal{G}(AA\mathcal{F}_1, C)$ if and only if $\forall S \in SCCS_{AA\mathcal{F}_1}$:

$$(E \cap S) \in \mathcal{G}(AA\mathcal{F}_1 \downarrow_{UP_{AA\mathcal{F}_1}(S, E)}, U_{AA\mathcal{F}_1}(S, E) \cap C).$$

We have the following equivalent conditions:

$$\begin{aligned} & E \in \mathcal{G}(AA\mathcal{F}_1, C) \\ \text{iff } & E \in \mathcal{CE}(AA\mathcal{F}_2, C \cap \mathcal{A}') && \text{(Definition of } \mathcal{G}) \\ \text{iff } & \forall S \in SCCS_{AA\mathcal{F}_2}: (E \cap S) \in \mathcal{CE}(AA\mathcal{F}_2 \downarrow_{UP_{AA\mathcal{F}_2}(S, E)}, U_{AA\mathcal{F}_2}(S, E) \cap (C \cap \mathcal{A}')) && \text{(Proposition 44)} \\ \text{iff } & \forall S \in SCCS_{AA\mathcal{F}_1}: (E \cap S) \in \mathcal{CE}(AA\mathcal{F}_2 \downarrow_{UP_{AA\mathcal{F}_1}(S, E)}, U_{AA\mathcal{F}_1}(S, E) \cap (C \cap \mathcal{A}')) && \\ & (SCCS_{AA\mathcal{F}_1} = SCCS_{AA\mathcal{F}_2}, UP_{AA\mathcal{F}_1}(S, E) = UP_{AA\mathcal{F}_2}(S, E) \text{ and } U_{AA\mathcal{F}_1}(S, E) = U_{AA\mathcal{F}_2}(S, E)) && \\ \text{iff } & \forall S \in SCCS_{AA\mathcal{F}_1}: (E \cap S) \in \mathcal{G}(AA\mathcal{F}_1 \downarrow_{UP_{AA\mathcal{F}_1}(S, E)}, U_{AA\mathcal{F}_1}(S, E) \cap C) && \text{(Definition of } \mathcal{G}) \end{aligned}$$

As far as preferred and grounded semantics for OR are concerned, the proofs are similar and are based on Proposition 45 and 46 respectively. $\square$

11.8.2. Stable Semantics

Lemma 11.13. *Given an $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubset \rangle$ and a set of arguments $E \subseteq \mathcal{A}$, E is a stable extension of $OR(AAF)$ if and only if $\forall S \in SCCS_{AA\mathcal{F}}$, it is the case that $E \cap S$ is a stable extension of $OR(AAF \downarrow_{UP_{AA\mathcal{F}}(S, E)})$.*

Proof. We denote $OR(AAF)$ as $\langle \mathcal{A}', \rightarrow' \rangle$.

From left to right. We need to show that:

- (1) $E \cap S \subseteq \mathcal{A}' \cap UP_{AA\mathcal{F}}(S, E)$;
- (2) $E \cap S$ is conflict-free in $OR(AAF \downarrow_{UP_{AA\mathcal{F}}(S, E)})$;
- (3) $\forall a \in \mathcal{A}' \cap UP_{AA\mathcal{F}}(S, E)$: $a \notin E \cap S$ implies that $\exists b \in E \cap S$ s.t. b attacks a in $OR(AAF \downarrow_{UP_{AA\mathcal{F}}(S, E)})$.

For (1), since E is a stable extension of $OR(AAF)$, we have that $E \cap S \subseteq E \subseteq \mathcal{A}'$. It remains to be shown that $E \cap S \subseteq UP_{AAF}(S, E)$. Let $a \in E \cap S$. Since E is conflict-free in $\langle \mathcal{A}, \rightarrow \rangle$, there is no $b \in E \setminus S$ such that $b \rightarrow a$. Thus $a \in UP_{AAF}(S, E)$. For (2), it is obvious that $E \cap S$ is conflict-free in $OR(AAF \downarrow_{UP_{AAF}(S, E)})$ since otherwise E would not be conflict-free in $\langle \mathcal{A}, \rightarrow \rangle$. For (3), since $a \in \mathcal{A}' \setminus E$ and E is a stable extension of $OR(AAF)$, there must be a $b \in E$ such that $b \rightarrow a$. Since $a \in UP_{AAF}(S, E)$, it follows that $b \notin E \setminus S$. Thus $b \in E \cap S$. Since $b \rightarrow a$ and $b \in E \cap S \subseteq \mathcal{A}' \cap UP_{AAF}(S, E)$, we have that b attacks a in $OR(AAF \downarrow_{UP_{AAF}(S, E)})$.

From right to left. We need to show that:

- (1) $E \subseteq \mathcal{A}'$;
- (2) E is conflict-free in $OR(AAF)$;
- (3) $\forall a \in \mathcal{A}'$: $a \notin E$ implies that $\exists b \in E$ such that $b \rightarrow a$.

For (1), since $E \cap S \subseteq \mathcal{A}'$ for all $S \in SCCS_{AAF}$, it follows that $E \subseteq \mathcal{A}'$. For (2), suppose, toward a contradiction, that there are $a, b \in E$ such that $a \rightarrow b$. We distinguish between two cases: (a) If $SCCS_{AAF}(a) = SCCS_{AAF}(b) = S$, that implies that $E \cap S$ is not conflict-free in $OR(AAF \downarrow_{UP_{AAF}(S, E)})$, a contradiction! (b) If $SCCS_{AAF}(a) \neq SCCS_{AAF}(b)$, we denote $SCCS_{AAF}(b)$ as S . Thus $b \in E \cap S$ and $b \notin UP_{AAF}(S, E)$, contradicting that $E \cap S \subseteq \mathcal{A}' \cap UP_{AAF}(S, E)$. $\square$

Proposition 86. The stable semantics for OR satisfies Principle 4.

Proof. For any $AAF = \langle \mathcal{A}, \rightarrow, \mathcal{S}, \sqsubseteq \rangle$ and $C \subseteq \mathcal{A}$, let $\mathcal{G}(AAF, C)$ be the set of stable extensions in $OR(AAF)$. Thus, for any $E \subseteq \mathcal{A}$, E is a stable extension in $OR(AAF)$ iff $E \in \mathcal{A}(AAF, \mathcal{A})$. On the other hand, we have that:

$$\begin{aligned}
 & E \in \mathcal{G}(AAF, C) \\
 \text{iff } & E \text{ is a stable extension in } OR(AAF) && \text{(Definition of } \mathcal{G}) \\
 \text{iff } & \forall S \in SCCS_{AAF}: (E \cap S) \text{ is a stable extension of } OR(AAF \downarrow_{UP_{AAF}(S, E)}) && \text{(Lemma 11.13)} \\
 \text{iff } & \forall S \in SCCS_{AAF}: (E \cap S) \in \mathcal{G}(AAF \downarrow_{UP_{AAF}(S, E)}, U_{AAF}(S, E) \cap C) && \text{(Definition of } \mathcal{G}) \quad \square
 \end{aligned}$$